Explicit finite extraction from convolution profiles for matrix multiplication

Abstract

We give a finite construction of matrix multiplication algorithms from convolution relations whose limiting exponent is 9/4. A determinant filtration gives exact restrictions with several output lengths. Rational occupation flows prescribe their frequencies with total logarithmic auxiliary cost linear in the profile index before integer scaling. Integral circulation rounding preserves every tensor balance and loses at most a cubic additive term in the logarithmic profile potential. Threefold balancing, a Fourier sector separator, exact interpolation, and repeated use of one catalyst then give an exact relative multiplication scheme. We retain all direct-sum multiplicities and finite costs. An explicit parameter family has recursive exponent less than $9/4 + 900/\log a$ for $a \geq 2^{64}$, with $\log A \leq 181a^5\lceil\log_2 a\rceil^2$ for its recursive source side A. Thus a prescribed integer parameter choice gives, for every positive rational ε, a finite arithmetic algorithm of exponent below $9/4 + \varepsilon$. Coefficient descent also gives a terminating construction of a rational bilinear seed with that exponent allowance. We identify the least scalar profile and compare the rounded construction with an exact binary certificate whose denominators grow at least exponentially. The results concern finite mathematical construction; they give neither a fixed exponent-9/4 seed nor a practical algorithm.

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1 Introduction

A bound on the matrix multiplication exponent and a finite description of an algorithm are different outputs of a proof. An asymptotic tensor argument can establish that a suitable rank decomposition exists without specifying its dimension or coefficients. A finite extraction chooses the tensor maps and their parameters, then accounts for the operations needed to use them. This paper gives such an extraction for the convolution relations introduced in the preprint *An Upper Bound of* $9/4$ *for the Matrix Multiplication Exponent* [14].

That preprint separates tensor blocks sharing one leg, applies the separation to polynomial multiplication, and derives inequalities for a symmetrized convolution profile. Its diagonal growth leads, through a detecting-character argument, to the claimed bound $\omega \leq 9/4$. Here ω is the infimum of arithmetic exponents for square matrix multiplication. The final algorithmic step chooses a matrix size and an exact rank decomposition whose existence follows from the bound. The chosen decomposition is not specified.

We construct the finite maps directly and give quantitative parameter choices. The main difficulty is to preserve the exact multiplicities of all tensor factors while controlling the cost of separation, interpolation, and auxiliary tensors. We use multiway convolution relations, choose their frequencies through a finite absorbing walk, and round the resulting rational counts while preserving every resource balance. This avoids clearing potentially large denominators. The construction gives a family whose fully counted exponents tend to $9/4$, with an explicit bound on its recursive source dimension.

1.1 Main results

A rational bilinear seed is a finite list of rational linear forms whose products and recombinations compute a square matrix product.

Theorem 1.1. *For every positive rational ε, a terminating procedure produces an integer $d \geq 2$ and an exact rational bilinear algorithm for $d \times d$ matrix multiplication using fewer than $d^{9/4+\varepsilon}$ bilinear multiplications, each multiplying a linear form in the left input by a linear form in the right input. A fixed recursion on this seed multiplies $n \times n$ matrices over any field of characteristic zero in $O_\varepsilon(n^{9/4+\varepsilon})$ arithmetic operations.*

In particular, $\omega \leq 9/4$. A bound with any positive real allowance follows by choosing a smaller positive rational allowance. The procedure generates identities by specified finite maps; it does not search for solutions to rank equations. Theorem 7.7 and Corollary 7.9 prove this statement.

The quantitative result first takes the form of a relative scheme: m independent products of $B \times B$ matrices are reduced to r products of $A \times A$ matrices. All input transformations and output recombinations are specified linear maps. Pooling the subproblems at each recursive level gives the exponent $p = \log(r/m)/\log(B/A)$, with all linear work included when $p > 2$.

Theorem 1.2 (Quantitative relative construction)**.** *For every integer $a \geq 2^{64}$, a specified finite construction gives positive integers r, m, A, B, with $B > A$, and an exact relative scheme over an explicitly determined cyclotomic field. Put $\ell_a = \lceil \log_2 a \rceil$. Its recursive exponent and source dimension satisfy*

$$\frac{9}{4} < p = \frac{\log(r/m)}{\log(B/A)} < \frac{9}{4} + \frac{900}{\log a}, \qquad \log A \leq 181 a^5 \ell_a^2.$$

Theorem 7.5 proves these bounds. In particular, for a positive rational allowance ε, one may choose

$$\ell = \max\left\{64, \left\lceil \frac{1800}{\varepsilon} \right\rceil\right\}, \qquad a = 2^\ell.$$

The resulting relative scheme has exponent below $9/4 + \varepsilon$ and $\log A \leq 181\, 2^{5\ell}\ell^2$. These choices require only integer and rational computations. The constants are conservative; the statement provides a proved finite choice for every allowance.

The bound on A concerns the recursive source of the relative scheme. To obtain the rational seed in Theorem 1.1, we unroll a fixed scheme to scalar multiplications, restrict its coefficients from its fixed number field to $\mathbb{Q}$, and select a sufficiently large dimension. This last procedure terminates, but the displayed bound on A is not a bound on the seed dimension d, its coefficient sizes, or its construction time.

1.2 The construction and its contributions

The scalar starting point is the least positive symmetric profile satisfying normalization, discrete concavity, and shifted tripling:

$$F(m,n) = s_{\min(m,n)}\left(\max(m,n) + \frac{\min(m,n)-1}{2}\right), \qquad s_a = \prod_{h=1}^{a-1}\left(1 + \frac{1}{3h}\right).$$

We prove both the formula and its minimality. Its diagonal growth is $a^{4/3}$, and the limiting exponent associated with a fixed profile index is

$$\tau_a = \frac{3\log(2a-1)}{\log F(a,a)}.$$

Every τ_a exceeds $9/4$, and $\tau_a \to 9/4$. The diagonal certificate uses tripling only at the row index. The full least-profile theorem allows all shifted tripling inequalities.

To realize the profile relations, we apply the determinant filtration of binary forms to convolution multiplication. An auxiliary factor of dimension d gives d convolution blocks. In a tensor power, retaining equal filtration counts on the two affected legs removes every off-diagonal term exactly. The tripling map requires only the number of middle blocks to agree; the two outer blocks can be chosen freely. This larger type selection works for every positive integer tripling count and removes the need for divisibility by three.

The affine profile determines rational transition probabilities among the blocks. Their occupation counts solve finite rational linear systems and give the exact resource identities. A square-root drift estimate bounds both the total occupation and the logarithmic auxiliary cost by $O(a)$. This is the quantitative reason for using multiway relations: their larger steps keep the expected cost of traversing a row bounded independently of its length.

We next round the transition counts to integers within their edgewise floor and ceiling bounds, preserving the resource identities exactly. For any integer scale $K \geq 1$, the rounded entropy expression has the form

$$K \log F(a,a) - \mathcal{L}, \qquad 0 \leq \mathcal{L} \leq 4J_a, \qquad J_a = O(a^3).$$

The estimate remains valid when occupation counts are arbitrarily small or zero. Thus the scale can be chosen for its error bound, independently of the denominators of the exact occupations. Exact denominator clearing for a binary certificate is analyzed separately as a benchmark; its exponential scale is not a lower bound for the rounded construction.

Three cyclic leg permutations suffice when successive rows use opposite orientations. After separation, the common source and output factors form an explicit catalyst. We retain it, first make the relation exact by interpolation, and then reuse the exact relation. Pooling the recursive products removes a per-level rounding loss. The final cost includes every source copy, output copy, interpolation evaluation, auxiliary factor, and catalyst supply. The smaller threefold balance alone does not improve the exponent: at corresponding doubled tensor powers, its charged scalar counts agree with those of sixfold balance.

The numerical bound $9/4$ was already announced in OpenAI [14]. Compared with that argument, the contributions here are the direct finite extraction, the multiway convolution restrictions, the quantitative occupation and rounding analysis, and the resulting explicit source-size bound. The proof is self-contained at the level of tensor maps and does not use the preprint's detecting-character theorem or spectral appendix.

1.3 Relation to previous work

Strassen's recursive use of a finite bilinear identity established the link between small matrix algorithms and an arithmetic exponent below three [17]. Approximate bilinear algorithms, tensor powers, and interpolation provide the standard passage from approximate identities to exact algorithms [3]. Schönhage's asymptotic sum inequality relates simultaneous matrix products to the exponent [15]. An accessible account of finite interpolation and the sum inequality is given by Stothers [16, Theorem 7 and the preceding discussion]. Our root-of-unity averaging is an explicit form of interpolation, and our pooled recursion proves the needed conversion of simultaneous outputs directly.

The asymptotic spectrum supplies a general language for comparing tensor powers [6, 19]. Relative bilinear complexity is an established setting for reductions between matrix multiplication tensors [18]. Catalytic tensor comparisons retain an auxiliary tensor on both sides of a reduction [5].

We use these familiar forms of comparison, while specifying the maps and retaining the cost of supplying the catalyst.

The determinant filtration itself is also established. Maakestad gives a universal integral Clebsch–Gordan filtration for binary forms, together with an explicit basis for its determinant kernel [13, Section 2, Lemma 2.5, Theorem 2.8 and Corollary 2.9]. Our application identifies the induced convolution multiplication on its successive quotients and the exact restrictions obtained by fixing the filtration counts. We do not claim a new Clebsch–Gordan filtration. Likewise, the existence of an integer circulation between integral lower and upper bounds is a standard consequence of total unimodularity [9, Theorem 2]. The algorithmic return-edge reduction and fractional-cycle rounding also appear in Kang and Payor [10, Sections 2.2–3.1]. We use circulation rounding to conserve tensor multiplicities; the separate weighted entropy estimate controls the cost of this rounding for our occupation flow. Related relative-entropy bounds for optimal fixed-total probability approximations appear in Böcherer and Geiger [4, Eq. (42)]; our action totals may change under the globally conserved rounding. The probabilistic tools for the occupation estimates are the fundamental matrix of an absorbing chain and a Doob transform [8, 12]; the required uniform auxiliary-cost estimate is proved here.

Fixed-frequency restrictions, entropy estimates, and products over leg permutations are standard in tensor-power methods; Le Gall gives a detailed example [11]. Threefold cyclic symmetrization is also explicit in Christandl et al. [5, Section 3.3]; the alternating row signs below make it compatible with the convolution resource ledger. Recent refinements of the laser method use carefully chosen distributions and asymmetric tensor powers [1, 2, 7]. For numerical context, the SODA 2025 paper of Alman et al. [2] proves $\omega < 2.371339$, and the 2026 preprint of Dupont et al. [7] reports $\omega < 2.371177$. The $9/4$ bound announced in the October 2026 preprint [14] is substantially stronger. The present proof derives that bound from finite convolution maps, independently of its spectral conclusion. Interpolation, type counting, circulation integrality, symmetrization, and bilinear recursion remain classical ingredients of the construction.

The closest source for the shared-leg separator and the initial convolution relations is OpenAI [14], Proposition 3.1 and Section 4. We give their coordinate maps, use exact type restrictions, and reduce the separator's source count from $5M$ to $2M - 1$. The subsequent occupation flows, conserved types, and cost estimates concern this specific family of relations. No claim of a first general constructive conversion of asymptotic tensor bounds is intended.

Indeed, existence of a suitable complex rank decomposition already permits an eventual symbolic search: its defining equations have rational coefficients, and a complex solution can be replaced by an algebraic one. The purpose of the present construction is to generate identities through prescribed operations and to bound their finite resources. This gives mathematical control of the extraction without solving an unspecified rank-decomposition problem.

1.4 Scope and organization

The results concern exact arithmetic with a fixed positive exponent allowance. They give neither one fixed recursion with exponent exactly $9/4$ nor a numerical stability guarantee. The relative source-size bound is explicit but enormous, and no useful finite-size implementation is established. The finite examples illustrate the individual charges in the proof; they are not measurements of a matrix multiplication program.

Section 2 fixes the tensor conventions and arithmetic model. Section 3 proves the least-profile theorem, and Section 4 gives the convolution restrictions, separator, and threefold balancing. Sections 5 and 6 construct the rational occupations and round them with exact resource balances. Section 7 composes the maps, proves the pooled recursion, gives the quantitative parameter family,

and extracts rational seeds. Section 8 presents finite examples, and Section 9 states the consequences and limitations. The appendices give the exact binary certificate and its denominator bound, full composition formulas, and optional separator refinements.

2 Tensors, restrictions, and arithmetic algorithms

All logarithms are natural. All vector spaces are finite dimensional. We first work over $\mathbb{C}$; every coefficient used in the construction will be algebraic. A tensor is written as a trilinear form in three disjoint variable groups, called its legs. Its rank $\mathrm{R}(T)$ is the least number of products of three linear forms whose sum is T. We write $A \trianglerighteq B$ if substitutions that are linear on each leg transform A into B. Such a restriction gives a bilinear algorithm for B using a bilinear algorithm for A. If the substitutions depend on a nonzero parameter t, and the transformed tensor is a polynomial $B + tB_1 + \cdots + t^D B_D$, we write $A \trianglerighteq B$. The individual substitutions may be Laurent polynomials; only their combined tensor is required to have no negative powers. Every degeneration below is supplied with its substitutions and a degree bound. No general approximation theorem is needed to use it.

For a positive integer s, the notation sT means the full direct sum of s copies of T, with independent variables on all three legs. It does not mean scalar multiplication of the coefficients. Tensor products distribute over these direct sums. We abbreviate tensor products by juxtaposition when no confusion can arise. We never cancel a direct-sum multiplicity or a tensor factor from a restriction.

Lemma 2.1 (Composition rules used below)**.** *Suppose a triple of Laurent-polynomial linear substitutions $\Psi_i(t)$ transforms a tensor A_i into $G_i(t) = \sum_{d=0}^{D_i} t^d G_{i,d}$. Let P_i be a specified restriction with $P_i(C_i) = A_i$, and let R_i be a triple of linear maps on the whole target variable spaces. If P_i, R_i are independent of t, then*

$$R_i \Psi_i(t) P_i(C_i) = \sum_{d=0}^{D_i} t^d R_i(G_{i,d}).$$

Their tensor product has degree at most $\sum_i D_i$. Its constant coefficient is $\bigotimes_i R_i(G_{i,0})$. Permuting the three legs preserves these statements.

Proof. The first identity follows by substitution and linearity on the tensor space. In a tensor product of polynomials with nonnegative exponents, degrees add, and a term has degree zero exactly when every factor does. Permuting the legs only permutes the three substitution maps. □

The square matrix multiplication tensor is

$$\mathsf{M}_d = \sum_{i,j,k=0}^{d-1} x_{ij} y_{jk} z_{ki}.$$

Its third variables encode the output coordinates by duality. Let $B_X(d) = x \sum_{i=0}^{d-1} y_i z_i$, and define $B_Y(d)$ and $B_Z(d)$ by permuting the legs. Relabeling coordinates gives

$$\mathsf{M}_d \mathsf{M}_e \cong \mathsf{M}_{de}, \qquad B_X(d) B_Y(d) B_Z(d) \cong \mathsf{M}_d. \tag{1}$$

For example, the three paired indices in the second product can be taken as the row, inner, and column indices of matrix multiplication. Consequently the product of all six leg permutations of $B_X(d)$ is M_{d^2}.

The basic convolution tensor is

$$C(r,b) = \sum_{i=0}^{r-1} \sum_{j=0}^{b-1} x_i y_j z_{i+j}. \tag{2}$$

It represents multiplication of polynomials of degrees at most $r-1$ and $b-1$. Put $d = r+b-1$ and let ξ be a primitive d-th root of unity. Evaluation at ξ^ℓ gives the explicit rank decomposition

$$C(r,b) = \frac{1}{d} \sum_{\ell=0}^{d-1} \left(\sum_i \xi^{\ell i} x_i \right) \left(\sum_j \xi^{\ell j} y_j \right) \left(\sum_{h=0}^{d-1} \xi^{-\ell h} z_h \right). \tag{3}$$

Indeed $0 \le i+j \le d-1$, so the Fourier sum selects exactly $h = i+j$. The output flattening has rank d, and hence $\mathrm{R}(C(r,b)) = d$.

For $\pi \in S_3$, superscript π denotes a permutation of the legs. Let $\mathcal{A}_+$ and $\mathcal{A}_-$ be the even and odd permutations, respectively. Define

$$S^\sigma_{r,b} = \bigotimes_{\pi \in \mathcal{A}_\sigma} C(r,b)^\pi, \qquad \sigma \in \{+,-\}. \tag{4}$$

The three factors remain labeled even when some are isomorphic. The following identities are legwise coordinate permutations:

$$S^\sigma_{r,b} \cong S^{-\sigma}_{b,r}, \qquad S^+_{r,r} \cong S^-_{r,r}, \qquad S^\sigma_{1,b} \cong \mathsf{M}_b, \qquad \mathrm{R}(S^\sigma_{r,b}) \le (r+b-1)^3. \tag{5}$$

Indeed swapping the inputs of $C(r,b)$ gives $C(b,r)$, and this odd permutation exchanges the two cosets. On the diagonal it also gives the second identity. Since $C(1,b) = B_X(b)$, the three factors in either coset put its exceptional leg in each global position once; (1) gives the third identity. The rank bound tensors the displayed evaluation decompositions. These isomorphisms do not exchange global legs of selected factors after they have been multiplied; they identify coordinates within each global leg.

For comparison with the exact binary certificate in Appendix A, put

$$Q_{r,b} = S^+_{r,b} S^-_{r,b} = \bigotimes_{\pi \in S_3} C(r,b)^\pi. \tag{6}$$

Then

$$Q_{r,b} \cong Q_{b,r}, \qquad \mathrm{R}(Q_{r,b}) \le (r+b-1)^6, \qquad Q_{1,2} \cong \mathsf{M}_4. \tag{7}$$

Arithmetic complexity counts scalar additions, subtractions, and multiplications, including multiplication by fixed coefficients. The algorithm may depend on a fixed exponent allowance ε, but not on the input dimension n. Its constant preprocessing cost may depend on ε. This is an arithmetic-operation statement, not a bit-complexity or floating-point stability statement.

Lemma 2.2 (Exactification)**.** *Suppose specified substitutions transform A into a polynomial $G(t) = \sum_{h=0}^{D} t^h G_h$. For every positive integer k,*

$$(kD+1) A^{\otimes k} \succeq G_0^{\otimes k}.$$

Proof. Set $E = kD+1$ and take a primitive E-th root η. The tensor polynomial $G(t)^{\otimes k}$ has degree at most kD. Thus

$$\frac{1}{E} \sum_{h=0}^{E-1} G(\eta^h)^{\otimes k} = G_0^{\otimes k}.$$

Each evaluation uses one copy of $A^{\otimes k}$; identifying its output variable groups and absorbing $1/E$ into one leg gives the claimed restriction. Every evaluation point is nonzero, so Laurent substitutions cause no problem. □

3 The least scalar profile

We first isolate the scalar problem. The connection to actual tensors will be proved by direct maps in Sections 4–7. In particular, no assumption that the least scalar profile is a tensor character enters the construction.

Definition 3.1. A profile is a positive symmetric function $P : \mathbb{N}_{>0}^2 \to \mathbb{R}$ satisfying

$$P(1, b) = b, \tag{8}$$

$$2P(a, b) \geq P(a, b-1) + P(a, b+1) \quad (b \geq 2), \tag{9}$$

$$P(a, 3h + a - 1) \geq 3P(a, h) \quad (h \geq 1). \tag{10}$$

Define

$$s_m = \prod_{h=1}^{m-1} \left(1 + \frac{1}{3h}\right), \qquad F(m, n) = s_{\min(m,n)} \left(\max(m, n) + \frac{\min(m, n) - 1}{2}\right). \tag{11}$$

An empty product equals one.

Theorem 3.2 (Least profile)**.** *The function F is a profile, and every profile P satisfies $P(m, n) \geq F(m, n)$. Moreover,*

$$F(m, n) \leq (m + n - 1)^{4/3}, \qquad \log F(a, a) = \frac{4}{3} \log a + O(1). \tag{12}$$

The first inequality is strict when $m = n > 1$.

Proof. Normalization and symmetry are immediate. Using $s_{b+1}/s_b = (3b + 1)/(3b)$ and $s_{b-1}/s_b = (3b - 3)/(3b - 2)$, direct subtraction gives

$$2F(a, b) - F(a, b-1) - F(a, b+1) = \begin{cases} \dfrac{2(a-b)s_b}{3b(3b-2)}, & 2 \leq b \leq a, \\ 0, & b > a. \end{cases}$$

Thus each row is concave. Its affine tail has slope s_a, so its extension bounds the row from above: $F(a, h) \leq s_a(h + (a - 1)/2)$. Evaluating this affine tail at $3h + a - 1 \geq a$ proves (10) for F.

For minimality, write $D_a = P(a, a)$, $k_a = 2D_a/(3a - 1)$, and $\Delta_{a,h} = P(a, h + 1) - P(a, h)$. Tripling at $h = a$ and decreasing increments imply

$$2D_a \leq P(a, 4a - 1) - P(a, a) \leq (3a - 1)\Delta_{a,a}.$$

By symmetry and concavity, for $a \geq 2$,

$$D_a - D_{a-1} = \Delta_{a-1,a-1} + \Delta_{a,a-1} \geq k_{a-1} + k_a.$$

It follows that $k_a \geq (3a - 2)k_{a-1}/(3a - 3)$. Since $k_1 = 1$, this proves $k_a \geq s_a$ and the diagonal bound.

Fix $m < n$. Starting with $x_0 = m$, put $x_{j+1} = 3x_j + m - 1$, and choose j with $x_j \geq n$. Then

$$x_j - m = (3^j - 1)(3m - 1)/2, \qquad P(m, x_j) \geq 3^j D_m.$$

A concave sequence lies above the chord joining two of its entries. The chord from m to x_j therefore gives

$$P(m, n) \geq D_m + \frac{n - m}{x_j - m}(3^j - 1)D_m = \frac{2D_m}{3m - 1}\left(n + \frac{m - 1}{2}\right) \geq F(m, n).$$

Only finitely many profile inequalities have been used.

Induction gives $s_a^3 \le (3a-1)/2$, strictly for $a > 1$. The induction step reduces to $(3a+2)(3a)^3 - (3a-1)(3a+1)^3 = 6a+1 > 0$. For $m \le n$, both $(3m-1)/2$ and $n + (m-1)/2$ are at most $m+n-1$. Cubing (11) proves (12) and its strict diagonal version. Finally, $\log(1+x) = x + O(x^2)$ and convergence of $\sum h^{-2}$ give $\log s_a = \frac{1}{3}\log a + O(1)$. □

The theorem makes precise a limitation of the scalar argument: these axioms, even together with $P(a,b) \le (a+b-1)^{4/3}$, cannot force diagonal growth faster than $a^{4/3}$. This is a statement about the scalar relaxation, not a claim that F comes from a tensor character. The off-diagonal minimality uses shifted tripling instances with h beyond the row index. The diagonal bound and the exact certificate in Appendix A need only the instances with h equal to the row index, which are the instances realized by our tensor maps.

The affine rows of F will determine transition probabilities for the finite construction. The scalar theorem identifies the best possible diagonal growth under the stated axioms; no tensor character attaining F is assumed.

4 Exact type restrictions and separation of a shared leg

We realize the profile relations by finite tensor maps. Each local map first produces matched sectors on two legs, with the third leg still shared. A Fourier separator then makes the sectors independent on all three legs. More precisely, the intermediate tensor is $\sum_\omega A_\omega$, where $A_\omega \in X \otimes Y_\omega \otimes Z_\omega$, the spaces Y_ω and Z_ω are direct-sum blocks, and X is common to all sectors. A separate change of first-leg coordinates in each sector is valid only after separation.

4.1 A determinant filtration

The filtration used here is the integral Clebsch–Gordan filtration; see Maakestad [13, Lemma 2.5, Theorem 2.8, and Corollary 2.9]. We give its bases to identify the convolution slices and then obtain an exact restriction by matching types in tensor powers.

Let $V_e = \mathbb{Q}[u,v]_e$ be the homogeneous binary forms of degree e. For integers $d \ge h \ge 1$, put

$$W_{d,h} = \mathbb{Q}[u,v]_d \otimes \mathbb{Q}[s,t]_h, \qquad \Delta = ut - vs.$$

Multiplication by a form in V_{r-1}, leaving the auxiliary variables s,t unchanged, represents $C(r, d+1)B_X(h+1)$. Its second-input and output spaces have the filtrations

$$F^j W_{d,h} = \Delta^j W_{d-j,h-j} \quad (0 \le j \le h), \qquad F^{h+1} W_{d,h} = 0.$$

After removal of Δ^j, diagonal substitution $(s,t) = (u,v)$ identifies

$$F^j W_{d,h} / F^{j+1} W_{d,h} \cong V_{d+h-2j}. \tag{13}$$

Here and below these are fixed-coordinate polynomial spaces; no representation-theoretic identification is needed.

We use an explicit integer basis for this filtration. Define

$$L_{p,q}(i) = \begin{cases} u^{p-i} v^i s^q, & 0 \le i \le p, \\ v^p s^{q-(i-p)} t^{i-p}, & p < i \le p+q, \end{cases}$$

and take

$$B_{j,i} = \Delta^j L_{d-j,h-j}(i), \qquad 0 \le j \le h, \quad 0 \le i \le d+h-2j. \tag{14}$$

To verify both the basis and (13), repeatedly replace ut by $vs + \Delta$. Modulo Δ, each monomial reduces to the unique displayed lift having the same total exponent of v, t. The remainder is divisible by Δ; repeat in bidegree $(d-1, h-1)$. The quotient monomials at each stage are independent under diagonal substitution. This gives unique integer coordinates and terminates after at most h such steps. The number of basis vectors is

$$\sum_{j=0}^{h}(d+h-2j+1) = (d+1)(h+1).$$

Use the same construction in the output space $W_{d+r-1,h}$. The third tensor leg uses the dual of this output basis; applying the input basis matrix to that leg would give the wrong tensor map.

Multiplication commutes with Δ, so it never decreases the filtration index. On quotient j, multiplication by $u^{r-1-\alpha}v^{\alpha}$ sends monomial index i to $i+\alpha$ with coefficient one. With $d = b-1$, the diagonal blocks are therefore exactly

$$C(r, b+h),\ C(r, b+h-2),\ \ldots,\ C(r, b-h).$$

Lemma 4.1 (Exact multiway types)**.** *Let $r \geq 1$, $h \geq 1$, and $b \geq h+1$. For nonnegative integers $n_0, \ldots, n_h$ with positive sum n, a specified restriction of $(C(r,b)B_X(h+1))^{\otimes n}$ has*

$$M = \frac{n!}{\prod_{j=0}^{h} n_j!}$$

matched sectors on its second and third legs. Each sector is isomorphic to $\prod_{j=0}^{h} C(r, b+h-2j)^{\otimes n_j}$. The first leg is shared; the sector-specific reordering of its tensor positions is made only after separation.

Proof. Use (14) in each tensor position. Retain on both affected legs only the words containing n_j occurrences of filtration index j. Every supported term has input indices j_ν and output indices j'_ν with $j'_\nu \geq j_\nu$ in each position ν. Both retained words have total index $\sum_j jn_j$, so every inequality is an equality. Thus all off-diagonal blocks disappear exactly. The remaining words agree positionwise; their number is the displayed multinomial coefficient. Their diagonal blocks give the claimed factors. □

For $h = 1$, the two quotient blocks are $C(r, b+1)$ and $C(r, b-1)$, the neighboring-length relation. For larger h, the auxiliary dimension is $h+1$, whereas h repeated binary steps consume dimension 2^h. This changes the finite tensor cost. It does not strengthen scalar concavity: pairing lengths equally far from b already gives $(h+1)P(r,b) \geq \sum_{j=0}^{h} P(r, b+h-2j)$. The exact type restriction itself has no interpolation degree.

4.2 Tripling with only the middle count fixed

Lemma 4.2 (Exact tripling types)**.** *Let $r \geq 2$, $n \geq 1$, and $0 \leq m \leq n$. A coordinate restriction of $C(r, 4r-1)^{\otimes n}$ has*

$$M_{\mathrm{t}}(n, m) = \binom{n}{m} 2^{n-m}$$

matched sectors on its second and third legs. Each sector contains $n-m$ factors isomorphic to $C(r,r)$ and m factors isomorphic to $C(r,r)^{(23)}$. The first leg remains shared. The choice $m = \lfloor (n+1)/3 \rfloor$ maximizes this count and gives

$$M_{\mathrm{t}}(n, m) \geq \frac{3^n}{n+1}. \tag{15}$$

No divisibility condition on n is required.

Proof. A supported term has $0 \le i < r$, $0 \le y < 4r-1$, and $z = i + y$. Partition the index sets into the following intervals, with endpoints included:

	left	middle	right
Y	$[0, r-1]$	$[r, 3r-2]$	$[3r-1, 4r-2]$
Z	$[0, 2r-2]$	$[2r-1, 3r-2]$	$[3r-1, 5r-3]$.

A left Y-index forces a left Z-index, and a right Y-index forces a right Z-index. Every mismatched term has a middle Y-label and an outer Z-label. Retain the words with exactly m middle labels on each leg, leaving their outer labels unrestricted. The number of middle labels cannot increase from input to output. Equality of their totals therefore removes every mismatched term. The surviving labels agree in every position, giving $\binom{n}{m}$ choices of middle positions and 2^{n-m} choices of outer labels.

Each outer block is a translate of $C(r, r)$. The middle block has terms

$$x_{r-1-i}\, y_{r+i+j}\, z_{2r-1+j}, \qquad 0 \le i, j < r.$$

Reversing the first index and shifting the interval coordinates identifies it with $C(r, r)^{(23)}$. This identifies a tensor with permuted leg dimensions; it does not perform a leg exchange on just one part of a tensor. The word-specific first-leg reversal is deferred until separation.

Finally, $M_{\mathrm{t}}(n, m+1)/M_{\mathrm{t}}(n, m) = (n-m)/(2(m+1))$. These ratios show that $\lfloor (n+1)/3 \rfloor$ is a maximizing choice, with a possible tie. The $n+1$ counts sum to 3^n, which proves (15). □

4.3 A Fourier separator

The following form includes an auxiliary dimension and a phase slope. The main construction uses only $K_0 = M$, $c = 1$.

Proposition 4.3 (Shared-leg separator)**.** *Let $A = \sum_{h=0}^{M-1} A_h$, with $A_h \in X \otimes Y_h \otimes Z_h$, where the Y_h and Z_h are pairwise disjoint sectors. For positive integers K_0, c, put $L = K_0 + c(M-1)$. There is an explicit degeneration*

$$LA \trianglerighteq \bigoplus_{h=0}^{M-1} (A_h B_X(K_0)). \tag{16}$$

Its polynomial has exponent support contained in $\{cd^2 : 0 \le d \le d_*\}$, where $d_* = \min(M-1, \lfloor (K_0 - 1)/c \rfloor)$. In particular, $K_0 = M, c = 1$ uses $2M-1$ source copies and has degree at most $(M-1)^2$. Permuting all three legs gives the same statement with any chosen shared leg.

Proof. Choose a primitive L-th root ζ. In source copy ℓ, $0 \le \ell < L$, make the substitutions

$$x_a^{(\ell)} \longmapsto \sum_{g=0}^{M-1} \zeta^{\ell c g} X_{a,g},$$

$$y_{h,b}^{(\ell)} \longmapsto \sum_{u=0}^{K_0-1} \zeta^{\ell(u-ch)} Y_{h,b,u},$$

$$z_{h,d}^{(\ell)} \longmapsto L^{-1} \sum_{v=0}^{K_0-1} \zeta^{-\ell v} Z_{h,d,v}.$$

Summing the copies retains exactly $u - v + c(g-h) \equiv 0 \pmod{L}$. Its absolute value is at most $L-1$, so the congruence is the equality $u - v + c(g-h) = 0$. Multiply the new variables by the respective powers of t

$$t^{cg^2}, \qquad t^{2hu-ch^2}, \qquad t^{-2hv}.$$

Every surviving term has total exponent $cg^2 + 2h(u-v) - ch^2 = c(g-h)^2$. It is zero exactly when $g = h$ and $u = v$. These terms are precisely the target in (16), whose first variables $X_{a,h}$ are now independent for different h. Moreover $c|g-h| = |u-v| \le K_0 - 1$, proving the exponent support. □

4.4 Threefold balancing

Products over the three cyclic leg permutations are an established balancing operation; see Christandl et al. [5, Section 3.3]. We use the two parity classes of permutations to make this operation compatible with the row boundaries of the convolution construction.

Let $\mathcal{A}_+$ and $\mathcal{A}_-$ be the even and odd permutations of three legs, as in the definition of $S^{\sigma}_{r,b}$. Apply a relation and its separator in each of the three orientations $\pi \in \mathcal{A}_\sigma$. The shared leg is the image of the native first leg under π; it occupies each global leg exactly once. Consequently the source auxiliary of a width-h relation becomes

$$B_X(h+1)B_Y(h+1)B_Z(h+1) \cong \mathsf{M}_{h+1},$$

and its separated convolution factors combine into the required $S^{\sigma}_{r,b+h-2j}$. The three separator auxiliaries likewise combine into M_M.

For tripling, put $C = C(r,r)$ and $\tau = (23)$. After separation, every retained word is put into the same order: the $n-m$ outer positions, followed by the m middle positions. Thus independent sector choices in the three orientations all have the same canonical sector tensor $C^{\otimes(n-m)}(C^{\tau})^{\otimes m}$. Regrouping these factors gives

$$\bigotimes_{\pi\in\mathcal{A}_\sigma} \left(C^{\otimes(n-m)}(C^{\tau})^{\otimes m}\right)^{\pi} \cong (S^{\sigma}_{r,r})^{\otimes(n-m)}(S^{-\sigma}_{r,r})^{\otimes m} \cong (S^{\sigma}_{r,r})^{\otimes n}. \tag{17}$$

The first isomorphism reorders tensor factors within each global leg; the second uses the diagonal sign isomorphism $S^{+}_{r,r} \cong S^{-}_{r,r}$. All sector-specific reversals have already been made on independent first-leg blocks. No individual factor undergoes an illicit exchange of two global legs.

Adjacent rows of the profile construction will use opposite signs. This makes their boundary factors agree under the actual isomorphism $S^{\sigma}_{r,b} \cong S^{-\sigma}_{b,r}$; using one sign for all rows would not give that identification. The signed resource accounting is carried out in the construction below. Appendix B specifies the complete relation maps, including their positive-degree terms. Further separator refinements appear in Appendix C; they are not needed for the main theorem.

5 Rational occupation flows with bounded auxiliary cost

We now choose how often to use each multiway relation. The choice must conserve all convolution factors and keep the product of the auxiliary dimensions small. Both requirements can be expressed through the occupation counts of a finite absorbing walk. Expected visits are obtained from the fundamental matrix of an absorbing chain [8, Theorem 11.4]; the change of probabilities below is a Doob transform by a positive harmonic function [12, Section 17.6.1]. The bounds on this particular walk will be proved directly. The probabilities below are prescribed by the least profile; no optimization oracle is needed.

Fix a target index $a \ge 2$. In row $2 \le r \le a$, put

$$n_r = 3r, \qquad x = b - r + 1, \qquad f_r(x) = \frac{3(r-1)}{2} + x \quad (0 \le x \le n_r).$$

The profile formula extends to the left boundary:

$$F(r, r-1+x) = s_r f_r(x) \qquad (0 \le x \le n_r). \tag{18}$$

For $x \geq 1$, this is (11). At $x = 0$, symmetry and $s_r/s_{r-1} = (3r-2)/(3r-3)$ give $F(r, r-1) = s_{r-1}(3r-2)/2 = 3(r-1)s_r/2$.

For an interior state $1 \leq x < n_r$, define

$$h_{r,x} = \min(x, n_r - x), \qquad d_{r,x} = h_{r,x} + 1,$$
$$\mathcal{Y}_{r,x} = \{x - h_{r,x}, x - h_{r,x} + 2, \ldots, x + h_{r,x}\}, \qquad q_{r,x,y} = \frac{f_r(y)}{d_{r,x} f_r(x)} \quad (y \in \mathcal{Y}_{r,x}). \tag{19}$$

The probabilities are positive rationals and sum to one, since the average of the children is x and f_r is affine. The determinant relation of Lemma 4.1 uses auxiliary dimension $d_{r,x}$ and these child columns. Its filtration index corresponding to child y is $(x + h_{r,x} - y)/2$. Its hypothesis holds: $h_{r,x} \leq x$ implies $b = r - 1 + x \geq h_{r,x} + 1$. Thus every child remains in the row interval $[r-1, 4r-1]$.

State zero absorbs. At state n_r, the walk returns deterministically to state one, representing the tripling relation from column $4r-1$ to column r. Inject one unit at

$$i_r = \begin{cases} 1, & r = a, \\ 2, & r < a. \end{cases}$$

Let $e_0, \ldots, e_{n_r}$ be the standard coordinate vectors for the states of this row.

Proposition 5.1 (Rational occupation flows)**.** *The walk is absorbed at zero in finite expected time, from every state. Its expected interior visit counts $\lambda_{r,x}$ and expected reset count μ_r are nonnegative rationals, determined by a nonsingular finite rational linear system. They satisfy the exact row identity*

$$\sum_{x=1}^{n_r-1} \lambda_{r,x} \left(e_x - \sum_{y \in \mathcal{Y}_{r,x}} q_{r,x,y} e_y \right) + \mu_r (e_{n_r} - e_1) = e_{i_r} - e_0. \tag{20}$$

The reset counts are

$$\mu_r = \begin{cases} \dfrac{1}{r-1}, & r = a, \\ \dfrac{2(3r-1)}{(r-1)(3r+1)}, & r < a. \end{cases} \tag{21}$$

In particular, $\mu_r \leq 2$. Each of the two row sums $\sum_x \lambda_{r,x}$ and $\sum_x \lambda_{r,x} \log d_{r,x}$ is less than 105 in the final row and less than 180 in a lower row. Consequently, with $C_a = 105 + 180(a-2)$,

$$\sum_{r=2}^{a} \sum_{x=1}^{n_r-1} \lambda_{r,x} \leq C_a, \qquad \sum_{r=2}^{a} \sum_{x=1}^{n_r-1} \lambda_{r,x} \log d_{r,x} \leq C_a. \tag{22}$$

In particular, every action satisfies $\lambda_{r,x} \log d_{r,x} \leq 180$. Unreachable states may have zero occupation.

Proof. Fix a row and suppress its subscript. First consider the walk that chooses each child in $\mathcal{Y}_x$ with equal probability $1/(h(x)+1)$, and stops at either endpoint. Write τ for its first endpoint visit. This walk is a martingale. The nonnegative function

$$g(x) = \sqrt{x} + \sqrt{n-x} - \sqrt{n}, \qquad \gamma = 1 - \frac{2\sqrt{2}}{3} > 0$$

has the drift bound

$$g(x) - \mathbb{E}_0[g(X_{t+1}) \mid X_t = x] \geq \gamma \sqrt{h(x)} \qquad (1 \leq x < n). \tag{23}$$

Here $\mathbb{E}_0$ denotes expectation for the uniform walk. To prove the bound when $x \le n/2$, its children are $0, 2, \dots, 2x$. Concavity of the square root, integrated on each unit interval, gives

$$\sum_{k=0}^{x} \sqrt{k} \le \frac{2}{3} x^{3/2} + \frac{1}{2}\sqrt{x}.$$

It follows that $\mathbb{E}_0\sqrt{X_{t+1}} \le (2\sqrt{2}/3)\sqrt{x}$. Jensen's inequality gives $\mathbb{E}_0\sqrt{n - X_{t+1}} \le \sqrt{n-x}$. These two inequalities prove (23); reflection about $n/2$ proves the other half.

Stop this estimate at $\tau \wedge T$, where T is a positive integer. Summing the conditional drift and using $g \ge 0$ gives

$$\gamma\, \mathbb{E}_{0,i} \sum_{t<\tau\wedge T} \sqrt{h(X_t)} \le g(i).$$

Monotone convergence therefore proves

$$\mathbb{E}_{0,i} \sum_{t<\tau} \sqrt{h(X_t)} \le \frac{g(i)}{\gamma} \le \frac{\sqrt{i}}{\gamma}. \tag{24}$$

Since $h(x) \ge 1$ in the interior, this also proves finite expected absorption, without assuming it in the stopping argument. The inequality $\log(1+h) \le \sqrt{h}$ for $h \ge 1$ follows because $\sqrt{h} - \log(1+h)$ is positive at one and has nonnegative derivative. Thus (24) bounds both the number of steps and the sum of $\log(h+1)$ during an excursion.

Now use the probabilities q, still stopping at the first endpoint. They are the transform of the uniform probabilities by the positive harmonic function f. For a path from i to x_t, the likelihood ratio telescopes to $f(x_t)/f(i)$, including when the last state is an endpoint. For every interior entry $i \ge 1$,

$$\frac{f(x_t)}{f(i)} \le \frac{f(n)}{f(i)} \le \frac{f(n)}{f(1)} = 3.$$

Applying this bound at each time to nonnegative charges gives

$$C_i \le \frac{3\sqrt{i}}{\gamma}, \tag{25}$$

where C_i denotes either the expected number of interior visits or the expected sum of $\log(h+1)$ in one q-excursion. In particular, these excursions also end in finite expected time.

The uniform walk exits at n with probability i/n: apply the martingale identity at $\tau \wedge T$, then use bounded convergence. The endpoint likelihood ratio therefore gives the q-exit probability

$$p_i = \frac{i f(n)}{n f(i)}, \qquad p_1 = \frac{1}{r}, \qquad p_2 = \frac{2(3r-1)}{r(3r+1)} < \frac{2}{r}. \tag{26}$$

After an exit at n, the reset starts a new excursion at one. For either of the charges in (25), the total expectation from entry i is consequently

$$C_i + \frac{p_i C_1}{1 - p_1}.$$

For entry one this is at most

$$\frac{6}{\gamma} = 54 + 36\sqrt{2} < 105.$$

For entry two it is at most

$$\frac{3(\sqrt{2}+2)}{\gamma} = 90 + 63\sqrt{2} < 180.$$

The expected number of resets is $p_i/(1-p_1)$, which gives (21). The same formulas give finite expectations from every other interior entry; from state n, there is one initial reset followed by entry at one. Thus absorption has finite expected time from every state of the full walk.

Let P_{uv} be its one-step transition probability from u to v, for $u, v \in \{1, \dots, n\}$. This transient matrix includes the reset state n. Finite expected absorption implies that $\sum_{t\geq 0} P^t$ has finite entries and is the inverse of $I - P$. The occupation vector is therefore the unique solution of

$$(I - P^{\mathsf{T}})(\lambda_1, \dots, \lambda_{n-1}, \mu)^{\mathsf{T}} = (\mathbf{1}_{\{v=i\}})_{v=1}^{n}.$$

The matrix and right-hand side are rational, so this solution is rational and can be obtained by exact Gaussian elimination. To recover the coordinate at zero as well, sum the successive differences $e_{X_t} - e_{X_{t+1}}$ up to absorption and take expectations. Their sum is $e_i - e_0$, proving (20). The row estimates give (22), and nonnegativity gives the individual action bound. □

The bound (22) controls two different costs: the total relation count before integer scaling, and the logarithm of the source auxiliary dimension. Both grow at most linearly with the number of rows. There are nevertheless up to $3r - 1$ interior actions in row r, each with up to $\lfloor 3r/2 \rfloor + 1$ children. Thus the whole construction has $O(a^2)$ possible actions and $O(a^3)$ child counts. Their rounding and finite-type costs must also be controlled. The next section does this while preserving the row identities exactly, rather than clearing the occupation denominators.

6 Integer types with exact resource balances

The occupations in Proposition 5.1 are rational, but their denominators need not be small. We do not clear those denominators. Instead, we round the transition counts while preserving every vertex balance. The resulting entropy loss has a bound that is uniform even when some occupations are arbitrarily small or zero.

The following integrality fact is classical: it follows from the total unimodularity of a directed incidence matrix and the integer-bound theorem of Hoffman and Kruskal [9, Theorem 2]. The return-edge reduction and rounding by fractional-cycle cancellation are described by Kang and Payor [10, Sections 2.2–3.1]. We give a residual-flow construction to specify the rounded counts used in the tensor maps. The estimate needed for our construction is the uniform weighted entropy loss for the chosen occupation flow, proved below.

Lemma 6.1 (Rounding a circulation)**.** *Let ϕ_e be a nonnegative rational circulation on a finite directed multigraph. There is an effective construction of an integer circulation n_e satisfying*

$$\lfloor \phi_e \rfloor \leq n_e \leq \lceil \phi_e \rceil \quad \textit{for every edge } e.$$

Thus $|n_e - \phi_e| < 1$, and every edge with an integer target is unchanged. Self-loops and parallel or opposing edges are allowed.

Proof. Put $\ell_e = \lfloor \phi_e \rfloor$, and at each vertex set

$$b_v = \sum_{e \text{ entering } v} \ell_e - \sum_{e \text{ leaving } v} \ell_e.$$

The fractional parts $f_e = \phi_e - \ell_e$ have outgoing minus incoming balance b_v. Discard loops for this step, since they have zero incidence. Give each nonintegral non-loop edge capacity one, in its original direction. Add an edge from a new source to each vertex with $b_v > 0$, of capacity b_v, and an edge from each vertex with $b_v < 0$ to a new sink, of capacity $-b_v$. The fractional parts, together with the added edges at full capacity, give a feasible flow of value $B = \sum_{b_v>0} b_v$.

An augmenting-path algorithm starting from zero finds an integral flow of this value. Indeed all residual capacities are integers, so each augmentation increases the value by a positive integer. If it stopped at a value below B, the vertices reachable from the source would define a cut of capacity below B, contradicting the feasible fractional flow. Add the resulting zero-or-one value on each original edge to its floor. Keep integer-target edges and loops at their floors. The source and sink edges are saturated, so the increments have balance b_v, and the new original counts have balance zero. Distinct edges and their residual reverse arcs are kept separately; therefore parallel and opposing edges cause no identification. This proves the claim and gives an algorithm using exact rational and integer arithmetic. □

Fix $a \geq 2$ and any integer $K \geq 1$. Use the row notation of Section 5. In row r, assign target count $K\lambda_{r,x}q_{r,x,y}$ to the transition $x \to y$, and target count $K\mu_r$ to the reset $n_r \to 1$. Equation (20) says that their outgoing minus incoming balance is $K(e_{i_r} - e_0)$. Add a return edge $0 \to i_r$ of value K. The augmented counts are a circulation, and this return edge remains exactly K under Lemma 6.1. Remove it after rounding. Write the remaining integer counts as $n_{r,x,y}$ and z_r, and put

$$w_{r,x} = \sum_y n_{r,x,y}, \qquad U = \sum_{r,x} w_{r,x}, \qquad T = \sum_r z_r.$$

They obey the exact row identities

$$\sum_{x=1}^{n_r-1} \left(w_{r,x}e_x - \sum_y n_{r,x,y}e_y \right) + z_r(e_{n_r} - e_1) = K(e_{i_r} - e_0), \tag{27}$$

and the strict error bounds

$$|n_{r,x,y} - K\lambda_{r,x}q_{r,x,y}| < 1, \qquad |z_r - K\mu_r| < 1, \qquad |w_{r,x} - K\lambda_{r,x}| < d_{r,x}. \tag{28}$$

An edge with zero target stays zero. In particular, an action with $\lambda_{r,x} = 0$ has $w_{r,x} = 0$. Zero-weight actions are omitted when constructing tensor maps. No divisibility condition is imposed on z_r.

6.1 The signed resource ledger

Put $b_r(x) = r - 1 + x$ and $\sigma_r = (-1)^{a-r}$. We record convolution factors using formal symbols $[\sigma; r, b]$, with the identifications

$$[\sigma; r, b] = [-\sigma; b, r], \qquad [+; r, r] = [-; r, r], \qquad [\sigma; 1, b] = [\mathsf{M}_b].$$

These are the legwise isomorphisms for the threefold tensors $S^{\sigma}_{r,b}$. The symbols belong to a free abelian group after these identifications; they record multiplicities of tensor factors.

Proposition 6.2 (Exact rounded ledger)**.** *The rounded counts satisfy*

$$\sum_{r=2}^{a} \sum_{x=1}^{n_r-1} \left(w_{r,x}[\sigma_r; r, b_r(x)] - \sum_y n_{r,x,y}[\sigma_r; r, b_r(y)] \right) + \sum_{r=2}^{a} z_r \left([\sigma_r; r, 4r-1] - [\sigma_r; r, r] \right) = K[+; a, a] - K[\mathsf{M}_2]. \tag{29}$$

Let $i_{\sigma,r,b}$ and $o_{\sigma,r,b}$ be the total source and output multiplicities after choosing representatives for the identified symbols, and set $c_{\sigma,r,b} = \min(i_{\sigma,r,b}, o_{\sigma,r,b})$. Then

$$\mathcal{C} = \bigotimes_{\sigma,r,b} (S^{\sigma}_{r,b})^{\otimes c_{\sigma,r,b}}, \qquad \mathcal{C}_N = \mathcal{C}^{\otimes N}, \qquad \sum_{\sigma,r,b} c_{\sigma,r,b} = U + T - K. \tag{30}$$

Boundary factors in this product are interpreted as M_b.

Proof. Replace the vectors in (27) by the corresponding symbols. The final row, whose injection is at $i_a = 1$, leaves $K([+; a, a] - [+; a, a-1])$. A lower row has injection at $i_r = 2$ and leaves $K([\sigma_r; r, r+1] - [\sigma_r; r, r-1])$. Since $\sigma_{r+1} = -\sigma_r$, its positive boundary equals the negative boundary of row $r+1$. The sum therefore leaves $K[+; a, a] - K[\sigma_2; 2, 1]$, which is (29). Before removing common multiplicities, both the source and output totals are $U + T$. Their difference has just the two distinct residual symbols in (29), each of multiplicity K. The sum of the coordinatewise minima is thus $U + T - K$. □

This is bookkeeping for actual source and output products. The common factors in (30) remain on both sides of the eventual tensor relation. They are not canceled from a restriction. The auxiliary factors of the determinant relations are charged separately.

6.2 Entropy loss without a minimum occupation bound

For a probability vector p, let $H(p) = -\sum_y p_y \log p_y$, where $0 \log 0 = 0$. For positive q, write $D(p\|q) = \sum_{p_y>0} p_y \log(p_y/q_y)$. This quantity is nonnegative: applying $\log t \le t - 1$ to q_y/p_y and summing gives $-D(p\|q) \le 0$. For an active action put $p_{r,x,y} = n_{r,x,y}/w_{r,x}$. Inactive actions contribute zero to all the sums below.

For fixed-total probability approximation, Böcherer and Geiger [4, Section IV.D, Eq. (42)] bound the relative entropy of the minimizing M-type approximation by $1/(Mq_{\min})$. Here an M-type has probabilities that are integer multiples of $1/M$. Thus $q_{\min} \ge 1/(3d)$ gives weighted loss at most $3d$ for that optimum. The empirical probabilities produced by our flow rounding need not minimize relative entropy, and the action total w may differ from its ideal total $K\lambda$. We prove the required estimate directly, including actions whose rounded counts are all zero.

Proposition 6.3 (Uniform entropy loss)**.** *Define*

$$G = K \log 2 + \sum_{r,x:w_{r,x}>0} w_{r,x}(H(p_{r,x}) - \log d_{r,x}) + T \log 3,$$
$$\mathcal{L} = \sum_{r,x:w_{r,x}>0} w_{r,x} D(p_{r,x}\|q_{r,x}).$$

Then

$$G = K \log F(a, a) - \mathcal{L}, \qquad 0 \le \mathcal{L} \le 4J_a, \tag{31}$$

where the number of candidate transition coordinates is

$$J_a = \sum_{r=2}^{a} \sum_{x=1}^{3r-1} d_{r,x} = \sum_{r=2}^{a} \left(\left\lfloor \frac{9r^2}{4} \right\rfloor + 3r - 1 \right)$$
$$= \frac{12a^3 + 42a^2 + 12a - 65 + (-1)^a}{16} = O(a^3). \tag{32}$$

The bound holds for every positive integer K.

Proof. First we give a uniform lower bound for the harmonic probabilities. In one row write $n = 3r$, $c = 3(r-1)/2$, $f(x) = c + x$, $h = \min(x, n-x)$, and $d = h + 1$. The smallest child has $y = x - h$. For $x \le n/2$, the ratio $f(x-h)/f(x) = c/(c+x)$ decreases with x. For $x \ge n/2$, the ratio $(c + 2x - n)/(c + x)$ increases with x. Comparing both with the continuous midpoint, whether or not it is an integer, gives

$$q_{r,x,y} \ge \frac{c}{d(c + n/2)} = \frac{r-1}{(2r-1)d} \ge \frac{1}{3d}. \tag{33}$$

For an active action abbreviate $V = K\lambda_{r,x}$, $w = w_{r,x}$, and $p = p_{r,x}$. By (28), $|w - V| < d$, and hence, for each positive n_y,

$$\frac{p_y}{q_y} < \frac{V}{w} + \frac{1}{wq_y} \le 1 + \frac{4d}{w}.$$

It follows that

$$wD(p\|q) \le w\log(1 + 4d/w) \le 4d.$$

This argument divides only by the positive integer w, never by the occupation $\lambda_{r,x}$. If $w = 0$, the action's loss is zero. If $\lambda_{r,x} = 0$, its counts are all zero by construction. Summing over the candidate actions proves $\mathcal{L} \le 4J_a$. The expression for J_a follows by summing $\sum_{x=1}^{3r-1}(1 + \min(x, 3r - x)) = \lfloor 9r^2/4 \rfloor + 3r - 1$.

For the identity in (31), use $q_{r,x,y} = F(r, b_r(y))/(d_{r,x}F(r, b_r(x)))$. Each active action satisfies

$$\begin{aligned} &w_{r,x}\log F(r, b_r(x)) - \sum_y n_{r,x,y}\log F(r, b_r(y)) \\ &\quad = w_{r,x}(H(p_{r,x}) - \log d_{r,x}) + w_{r,x}D(p_{r,x}\|q_{r,x}). \end{aligned}$$

The same equality reads $0 = 0$ for an inactive action. Each reset contributes $z_r \log 3$, because $F(r, 4r-1) = 3F(r, r)$. Apply the symmetric potential $\log F$ to (29). Its total is $K\log F(a, a) - K\log 2$, proving the identity. The reset counts therefore cause no additional entropy error. □

The same rounding supplies the bounds needed to charge auxiliary dimensions and interpolation. Put $W = \sum_{r,x} w_{r,x}\log d_{r,x}$ and $C_a = 105 + 180(a-2)$. Proposition 5.1 bounds both $\sum_{r,x}\lambda_{r,x}\log d_{r,x}$ and $\sum_{r,x}\lambda_{r,x}$ by C_a. These two bounds and (28) give

$$\begin{aligned} W \le KC_a + J_a\log(2a), \qquad U \le KC_a + J_a, \\ T \le (2K+1)(a-1), \\ w_{r,x}\log d_{r,x} \le 180K + 2a\log(2a). \end{aligned} \tag{34}$$

Indeed $d_{r,x} \le \lfloor 3a/2 \rfloor + 1 \le 2a$, so summing $w_{r,x} < K\lambda_{r,x} + d_{r,x}$ proves the first two bounds. The exact reset occupations obey $\mu_r \le 2$, whence $z_r < 2K + 1$. Finally, the nonnegative ideal charge of an individual action is at most its row's charge, which is at most 180. Multiplying its rounding bound by $\log d_{r,x}$ proves the last inequality. This individual bound will control the largest sector count; the total bound for W alone would be weaker by a factor of order a.

7 The counted finite construction

Fix a profile index $a \ge 2$ and a positive integer rounding scale K. Use the integer counts of Proposition 6.2. Let $\mathcal{I}_c$ index the actions with $w_i > 0$, and let $\mathcal{I}_t$ index the resets with $z_r > 0$. Put $\mathcal{J} = \mathcal{I}_c \sqcup \mathcal{I}_t$ and $I = |\mathcal{J}|$. An action $i = (r, x)$ has auxiliary dimension $d_i = d_{r,x}$, child counts $n_{i,y}$, and empirical distribution $p_{i,y} = n_{i,y}/w_i$. Counts for children are assigned to the filtration index $(x + h_{r,x} - y)/2$.

The remaining parameters have distinct roles. A type scale N reduces the relative loss in counting sector words. A tensor power k amortizes interpolation, and the number j of exact reuses amortizes the catalyst. Unlike exact normalization of the binary certificate, the choice of K is unrestricted.

At scale N, define the sector counts

$$M_i = \frac{(Nw_i)!}{\prod_y (Nn_{i,y})!} \quad (i \in \mathcal{I}_c), \qquad M_r = \binom{Nz_r}{m_r} 2^{Nz_r - m_r}, \quad m_r = \left\lfloor \frac{Nz_r + 1}{3} \right\rfloor \quad (r \in \mathcal{I}_t). \tag{35}$$

Zero child counts contribute $0! = 1$. Put

$$M = \prod_{i \in \mathcal{J}} M_i, \qquad S = \prod_{i \in \mathcal{J}} (2M_i - 1)^3, \qquad H = M^3, \tag{36}$$

$$D = 3 \sum_{i \in \mathcal{J}} (M_i - 1)^2, \qquad u = \prod_{i \in \mathcal{I}_c} d_i^{Nw_i}, \qquad v = 2^{NK} M. \tag{37}$$

Here S is an integer multiplicity; the tensors $S^{\sigma}_{r,b}$ always carry indices. Let $\mathcal{C}_N$ be the catalyst in (30).

Proposition 7.1 (Counted catalytic degeneration)**.** *The specified maps give a polynomial degeneration of degree at most D:*

$$S\mathcal{C}_N (S^{+}_{a,a})^{\otimes NK} \mathsf{M}_u \trianglerighteq H\mathcal{C}_N \mathsf{M}_v. \tag{38}$$

Proof. For each action use the fixed-type restriction of Lemma 4.1, and for each reset use the partial-middle restriction of Lemma 4.2. In row r, take the three permutations in the coset with sign $\sigma_r = (-1)^{a-r}$. For each such permutation apply the separator of Proposition 4.3, with auxiliary dimension M_i. It consumes $2M_i - 1$ source copies and produces M_i independent copies of the sector tensor, each with a dot product of dimension M_i.

Write the type restriction as T_i, the separator map as $\mathsf{S}_i(t)$, and the sector-specific coordinate maps after separation as R_i. The complete map for a permuted relation is

$$\Psi_{i,\pi}(t) = \mathsf{R}_i^{\pi} \circ \mathsf{S}_i(t)^{\pi} \circ \left(\bigoplus_{\ell=0}^{2M_i - 2} \mathsf{T}_i^{\pi} \right). \tag{39}$$

For a reset, group its three permuted factors before using the diagonal identity in (5). This identifies every separated sector with the same power of $S^{\sigma_r}_{r,r}$, regardless of its outer labels. Appendix B specifies these maps on the whole target spaces, including positive-degree terms.

Tensoring all relation maps gives the copy counts S and H. The signed ledger (29) identifies the convolution factors on the two sides with $\mathcal{C}_N (S^{+}_{a,a})^{\otimes NK}$ and $\mathcal{C}_N \mathsf{M}_{2^{NK}}$, respectively. Each action contributes three dot products of dimension $d_i^{Nw_i}$, one in each leg. Their product is M_u. The separator outputs similarly contribute M_M, giving M_v. Use one common parameter for all separators. Their degrees add by Lemma 2.1, giving D. All preceding type maps and subsequent coordinate maps are independent of that parameter. □

The estimate $1 \le S/H < 2^{3I}$ bounds a cost. Neither direct-sum multiplicity has been canceled.

7.1 Exactification and repeated use of one catalyst

Define the explicit rank bound

$$\kappa_N = \prod_{\sigma,r,b} (r + b - 1)^{3Nc_{\sigma,r,b}} \ge \mathrm{R}(\mathcal{C}_N). \tag{40}$$

It tensors the evaluation decompositions (3). The boundary factor M_2 has zero catalyst multiplicity, so this product uses only the remaining convolution factors.

Proposition 7.2 (An exact relative multiplication scheme)**.** *For positive integers k, j, put $E = kD + 1$, $A = u^{kj}$, $B = v^{kj}$, and*

$$r_* = \kappa_N^k E^j S^{kj} (2a-1)^{3NKkj}, \qquad m_* = H^{kj}. \tag{41}$$

There is a specified exact restriction $r_* \mathsf{M}_A \succeq m_* \mathsf{M}_B$.

Proof. Apply Lemma 2.2 to the k-th power of (38). Replace each source copy of $S_{a,a}^{+}$ by its rank-$(2a-1)^3$ evaluation decomposition. Writing $\alpha = ES^k(2a-1)^{3NKk}$ and $\beta = H^k$, we obtain

$$\alpha \mathcal{C}_N^{\otimes k} \mathsf{M}_{u^k} \succeq \beta \mathcal{C}_N^{\otimes k} \mathsf{M}_{v^k}.$$

Apply this exact restriction j times while retaining the same catalyst:

$$\alpha^j \mathcal{C}_N^{\otimes k} \mathsf{M}_{u^{kj}} \succeq \beta^j \mathcal{C}_N^{\otimes k} \mathsf{M}_{v^{kj}}.$$

For the induction, tensor the preceding restriction with $\alpha \mathsf{M}_{u^k}$, then apply the one-step restriction to the retained catalyst in each output summand. The matrix factors combine by (1). The catalyst exponent stays k. Supply its source using κ_N^k rank-one tensors and restrict the output catalyst to a coefficient equal to one. Such a coefficient is available by choosing a monomial in every convolution factor. This proves (41). □

Exactification precedes reuse. Iterating the degeneration itself would require a separate bound on the accumulated approximation degree.

7.2 Pooled recursion with padding

Lemma 7.3 (Relative schemes give recursive algorithms)**.** *Suppose an explicit restriction $r\mathsf{M}_A \succeq m\mathsf{M}_B$ is given, with positive integers $B > A$. Set*

$$q = r/m, \qquad p = \frac{\log q}{\log(B/A)}.$$

Then $p \geq 2$. The restriction yields a fixed arithmetic algorithm with cost $O(n^p)$ if $p > 2$, and $O(n^2 \log n)$ if $p = 2$. Every scalar product is a product of a linear form in the left input entries and a linear form in the right input entries.

Proof. The first-leg flattenings have ranks rA^2 and mB^2. Rank cannot increase under restriction, so $q \geq (B/A)^2 > 1$.

Pool all products at a given recursion depth. If there are v_ℓ products of side n_ℓ, group them in batches of m, filling the last batch with zero products. Pad each input to side $B\lceil n_\ell/B \rceil$, divide it into a $B \times B$ array of blocks of side $\lceil n_\ell/B \rceil$, and apply the restriction. This gives

$$v_{\ell+1} = r\lceil v_\ell/m \rceil \leq q v_\ell + r, \qquad n_{\ell+1} = A\lceil n_\ell/B \rceil.$$

The scalar identity remains valid on matrix blocks: its products place a left-input factor before a right-input factor, and its coefficients are central scalars. Starting with $v_0 = 1$, the count obeys $v_\ell \leq q^\ell(1 + r/(q-1))$. There is a constant C, depending only on the scheme, for which all transformations at this level take at most $C v_\ell n_\ell^2$ operations. To justify pooling across different

parents, perform the input transformations in increasing depth, store their linear output maps, and perform the output combinations in reverse depth after computing the leaves. Products at the same depth can then be grouped together without changing any algebraic dependency.

Put $\rho = A/B$ and $c = A/(1-\rho)$. Since $n_{\ell+1} \le \rho n_\ell + A$, $n_\ell \le c + \rho^\ell(n-c)$. Use classical multiplication at the fixed leaf threshold $n_* = \lceil c \rceil + 1$. There are at most $\log_{1/\rho} n + O(1)$ levels. The total work is bounded by a constant times the sum of $q^\ell(1+\rho^{2\ell}n^2)$ over these levels, including leaves. As $q\rho^2 \ge 1$, this gives the two asserted bounds. At every stage the input transformations are separately linear in the original left and right entries. This remains true at the classical leaves and proves the last assertion. □

For the relative scheme above, write $\Delta_N = \log(v/u)$. Whenever $\Delta_N > 0$, its pooled exponent is exactly

$$p = \frac{3NK\log(2a-1) + \log(S/H) + \log E/k + \log \kappa_N/j}{\Delta_N}. \tag{42}$$

For comparison with saved finite cost certificates, also define the slightly larger bound

$$\Phi = \frac{3NK\log(2a-1) + \log(S/H) + \log E/k + \log \kappa_N/j + \log 2/(kj)}{\Delta_N}. \tag{43}$$

To see the origin of the extra term, let $\mathrm{cost}(n)$ be the cost of one batch of m_* products when each parent's children are grouped separately. With $b_0 = \lceil r_*/m_* \rceil$, this schedule obeys

$$\mathrm{cost}(n) \le b_0\,\mathrm{cost}(A\lceil n/B \rceil) + O(n^2).$$

The dimension estimate in Lemma 7.3 gives exponent $\log b_0/\log(B/A)$, bounded by Φ since $b_0 \le 2r_*/m_*$. Pooling removes this last ceiling charge. Both expressions retain the actual copy, interpolation, and catalyst costs. The source M_A is the recursive operation; no fast decomposition of it has been assumed.

7.3 Type losses and finite cost bounds

Lemma 7.4 (A quantitative type bound)**.** *Let $n_1 + \cdots + n_s = n$, with all $n_i > 0$, and put $p_i = n_i/n$. Then*

$$\frac{\exp(nH(p))}{(n+1)^{s-1}} \le \frac{n!}{\prod_i n_i!} \le \exp(nH(p)), \qquad H(p) = -\sum_i p_i \log p_i.$$

Proof. The multinomial probability of the type (n_i) is its coefficient times $\exp(-nH(p))$, and is at most one. It is also a mode: the factor $n_i^h/h!$ is maximized at $h = n_i$ or $h = n_i - 1$, as follows by comparing consecutive terms. The joint probability, up to a common factor, is the product of these factors. There are at most $(n+1)^{s-1}$ types, so a mode has probability at least their reciprocal. □

For an action i, let $s_i = |\{y : n_{i,y} > 0\}|$ and $H_i = w_i H(p_i)$. For a reset put $H_r = z_r \log 3$. Define

$$\begin{aligned} \mathcal{T}_N &= \sum_{i \in \mathcal{I}_c} (s_i - 1)\log(Nw_i + 1) + \sum_{r \in \mathcal{I}_t} \log(Nz_r + 1), \\ B_N &= N \max_{i \in \mathcal{J}} H_i. \end{aligned} \tag{44}$$

Lemma 7.4 and the modal estimate for partial-middle tripling give

$$NG - \mathcal{T}_N \le \Delta_N \le NG, \qquad G = K\log F(a,a) - \mathcal{L}, \qquad 0 \le \mathcal{L} \le 4J_a. \tag{45}$$

Only the number of used children enters the type penalty. The source auxiliary dimension remains d_i, including when some children have count zero.

The other charges have the elementary bounds

$$\begin{aligned} 0 \le \log(S/H) &< 3I \log 2, \\ \log E &\le \log(3kI+1) + 2B_N, \\ \log \kappa_N &\le 3N(U+T-K)\log(5a). \end{aligned} \tag{46}$$

Indeed $M_i \le \exp(NH_i)$, and $E = 1 + 3k\sum_i (M_i - 1)^2 \le (3kI+1)\exp(2B_N)$. Every rank base $r+b-1$ is less than $5a$; the catalyst count is $U+T-K$. These bounds do not require expanding the sector words or writing out any tensor coefficients.

The uncharged profile expression

$$\tau_a = \frac{3\log(2a-1)}{\log F(a,a)} \longrightarrow \frac{9}{4} \tag{47}$$

remains useful for comparison. Since $\Delta_N \le NK \log F(a,a)$ and all numerator charges in (42) are nonnegative, every scheme here with $B > A$ has $p \ge \tau_a > 9/4$. The finite construction therefore approaches the endpoint from above.

7.4 An explicit rate of convergence

The preceding estimates give more than convergence to $9/4$. The following bounds specify an index for every exponent allowance. The constants are deliberately conservative.

Theorem 7.5 (A fully specified family)**.** *For an integer $a \ge 2^{64}$, choose*

$$\ell_a = \lceil \log_2 a \rceil, \qquad K = a^3 \ell_a, \qquad N = k = 1, \qquad j = a\ell_a.$$

Apply conserved rounding to the maximal-width occupations, and use the resulting integer types in the finite construction. The relative scheme has $B > A$. Its pooled recursive exponent p satisfies

$$p < \frac{9}{4} + \frac{900}{\log a}, \qquad \log A \le 181 a^5 \ell_a^2. \tag{48}$$

In particular, the complete finite construction has exponent tending to $9/4$ and $\log A = O(a^5 (\log a)^2)$.

Proof. Write $t = \log a$ and $\ell = \ell_a$. Then $t > 32$, $\ell \ge 64$, $t/\log 2 \le \ell \le a$, and $\log K \le 4t$. We first bound the contraction, including both rounding and type losses. The candidate coordinate count and the number of active relations obey $J_a \le 2a^3$ and $I \le 2a^2$. The occupation and rounding bounds give, for each action and reset,

$$w_i \le 180K + 2a, \qquad z_r \le 2K + 1.$$

Consequently the type loss at $N = 1$ is bounded by

$$\mathcal{T}_1 \le J_a \log(180K + 2a + 1) + (a-1)\log(2K+2). \tag{49}$$

Here the sum of the action support sizes minus one is at most J_a, and there are at most $a-1$ positive resets. Since $180K + 2a + 1 \le 181K$, $\log 181 < 6$, and $\log 4 < 2$,

$$\frac{4J_a + \mathcal{T}_1}{K} \le \frac{20 + 8t + (4t+2)/a^2}{\ell} \le \frac{\frac{41}{2} + 9t}{\ell} < \frac{41}{128} + \frac{63}{10}.$$

The first two inequalities use $(a-1)/a^3 \le 1/a^2 \le 1/4$; the last uses $\ell \ge 64$ and $t/\ell \le \log 2 < 7/10$.

There is also a convenient elementary lower bound for the profile. The inequality $\log(1+x) \ge x - x^2/2$, for $x \ge 0$, gives

$$\log s_a \ge \frac{1}{3}\sum_{h=1}^{a-1}\frac{1}{h} - \frac{1}{18}\sum_{h=1}^{a-1}\frac{1}{h^2} \ge \frac{1}{3}\log a - \frac{1}{9}.$$

As $F(a,a) = s_a(3a-1)/2 \ge a s_a$, we obtain $\log F(a,a) \ge (4/3)t - 1/9$. The conserved potential identity and the type bound therefore imply

$$\frac{\log(v/u)}{K} \ge \log F(a,a) - \frac{4J_a + \mathcal{T}_1}{K} > \frac{4}{3}t - 7 > 0, \tag{50}$$

where $1/9 + 41/128 + 63/10 < 7$. This proves $B > A$.

We next bound every term in the exponent numerator. The copy charge satisfies

$$\frac{\log(S/H)}{K} \le \frac{6}{a\ell} \le 1.$$

For interpolation, each action entropy is at most its auxiliary charge, and the per-row occupation estimate gives

$$H_i \le w_i \log d_i \le 180K + 2a\log(2a).$$

The reset entropy $z_r \log 3$ is smaller than the same upper bound. Thus, writing B_1 for the largest of these entropy bounds,

$$\frac{\log E}{K} \le \frac{\log(3I+1) + 2B_1}{K} \le 360 + \frac{5}{a\ell} \le 361.$$

In the middle inequality we used $\log(6a^2+1) \le 2a$ and $\log(2a) \le a$, valid for $a \ge 2$.

Finally, the rounded count bounds imply

$$U + T - K \le 182aK + 2a^3 + a < 184aK.$$

All catalyst rank bases are less than $5a$, so

$$\frac{\log \kappa_1}{jK} \le 552\frac{\log(5a)}{\ell} < 552 \cdot \frac{7}{10}\left(1 + \frac{2}{8}\right) = 483.$$

This uses $\log 2 < 7/10$, $\log 5 < 2$, and $t > 8$. Together with $3\log(2a-1) < 3t+3$, these estimates bound the pooled exponent numerator by $K(3t+850)$. This bound even allows the additional fixed-batch ceiling charge $\log 2/j$, which is less than K. Combining it with (50) gives

$$p < \frac{3t+850}{\frac{4}{3}t - 7} = \frac{9}{4} + \frac{850 + 63/4}{\frac{4}{3}t - 7} < \frac{9}{4} + \frac{900}{t},$$

because $(4/3)t - 7 \ge t$ for $t > 32$. Conversely, every overhead is nonnegative and $\log(v/u) \le K \log F(a,a)$, so

$$p \ge \frac{3\log(2a-1)}{\log F(a,a)} > \frac{9}{4}$$

by the strict diagonal bound in Theorem 3.2. The upper and lower bounds prove the claimed limit.

For the source dimension, use $C_a \le 180a$ and the rounded auxiliary bound to obtain

$$\log A = j \log u \le a\ell(180aK + 2a^3\log(2a)) = a^5\ell^2\left(180 + \frac{2\log(2a)}{a\ell}\right) \le 181a^5\ell^2.$$

The last inequality follows from $\log(2a) \le a$ and $\ell \ge 64$. □

Corollary 7.6 (An explicit index for an exponent allowance)**.** *Given a positive rational ε, put*

$$\ell = \max\left\{64, \left\lceil \frac{1800}{\varepsilon} \right\rceil\right\}, \qquad a = 2^{\ell}.$$

The family in Theorem 7.5 gives a finite relative scheme with recursive exponent strictly below $9/4 + \varepsilon$, and

$$\log A \le 181\, 2^{5\ell}\ell^2.$$

The index and all four construction parameters are selected by integer and rational computations.

Proof. Here $\ell_a = \ell$ and $\log a = \ell \log 2 > \ell/2$. Therefore $900/\log a < 1800/\ell \le \varepsilon$, and the source bound follows from (48). □

The bound concerns the recursive source dimension A of the relative scheme. It does not bound the side of a rational square seed obtained after coefficient descent, or the time and bit complexity needed to write out all coefficients of the construction.

7.5 An effective algorithm over a specified field

Theorem 7.7 (Effective finite extraction)**.** *There is a terminating procedure that, given a positive rational ε, specifies a finite arithmetic algorithm for multiplying arbitrary complex $n \times n$ matrices in $O_\varepsilon(n^{9/4+\varepsilon})$ operations. All coefficients lie in an explicitly specified cyclotomic field. The procedure uses only finite integer and rational computations, coordinate permutations, tensor products, and the maps proved above.*

Proof. Choose a, K, N, k, j by Corollary 7.6 and Theorem 7.5. Compute the rational occupations by their finite linear systems, round their circulations using the integral max-flow procedure in Lemma 6.1, and form the integer counts and catalyst. Proposition 7.2 gives the exact relative scheme. Theorem 7.5 and Lemma 7.3 give an exponent strictly below $9/4 + \varepsilon$. The possible quadratic-logarithmic term is also within this allowance.

For a deterministic specification, order states, edges, tensor positions, and sector words lexicographically, and use the first available augmenting path at each max-flow step. Use the displayed determinant bases, interval sectors, Fourier substitutions, evaluation decompositions, and root averages. Separate sectors before applying their individual coordinate maps. Compose the maps in the order in the proofs of Propositions 7.1 and 7.2. Every root of unity has an explicitly known finite order. One may take

$$L_* = \operatorname{lcm}\big(E,\, 2a-1,\, \{2M_i - 1 : i \in \mathcal{J}\},\, \{r + b - 1 : c_{\sigma,r,b} > 0\}\big).$$

All coefficients lie in $\mathbb{Q}(\zeta_{L_*})$, represented by the power basis modulo the cyclotomic polynomial. Catalyst reuse and recursion introduce no further roots of unity. Thus the procedure specifies finite coefficient arrays and their composition; expanding them is not needed for the proof of correctness. □

7.6 Rational coefficients and an explicit square seed

The field issue does not require an existence theorem for algebraic solutions of rank equations.

Lemma 7.8 (Restriction of coefficients to the rationals)**.** *Let T have rational coefficients, and suppose an explicit rank-R decomposition of T is given over a number field $\mathbb{K}$ of degree d. Then an explicit rational decomposition has length at most d^2R.*

Proof. Choose a rational basis $e_1 = 1, e_2, \dots, e_d$ of $\mathbb{K}$, and let $\phi : \mathbb{K} \to \mathbb{Q}$ extract the coefficient of e_1. For each summand write its first two linear forms as $\sum_s e_s \alpha_s$ and $\sum_t e_t \beta_t$, where the α_s, β_t have rational coefficients, and call its third form γ. Apply ϕ coefficientwise to the tensor identity. This summand becomes $\sum_{s,t} \alpha_s \beta_t \phi(e_s e_t \gamma)$, a sum of at most d^2 rational rank-one tensors. The target is unchanged because its coefficients are rational and $\phi(1) = 1$. All field operations are effective; for the field above, use the power basis modulo the cyclotomic polynomial. □

Corollary 7.9 (Finite rational seeds)**.** *For every positive rational ε, a terminating procedure produces an integer $d_0 \geq 2$ and a rational rank decomposition of M_{d_0} of length $R_{\text{seed}} < d_0^{9/4+\varepsilon}$. Consequently a fixed rational bilinear recursion attains an exponent strictly below $9/4 + \varepsilon$.*

Proof. Apply Theorem 7.7 with $\varepsilon/2$. For each integer dimension n, unroll its pooled recursion to classical leaves, and set the unused initial input pairs to zero. Lemma 7.3 gives a bilinear decomposition of M_n. Its number of leaf products is at most its arithmetic cost, hence is $O_\varepsilon(n^{9/4+\varepsilon/2})$. Its coefficient field is fixed independently of n. Apply Lemma 7.8; this multiplies the length by at most a fixed constant.

Enumerate $n = 2, 3, \dots$, generate the finite recursion and its rational decomposition, and stop when its length R_n satisfies $R_n^{\beta_\varepsilon} < n^{\alpha_\varepsilon}$, where the positive integers $\alpha_\varepsilon, \beta_\varepsilon$ satisfy $\alpha_\varepsilon/\beta_\varepsilon = 9/4+\varepsilon$. The strict gap in exponents proves termination, regardless of the fixed constants. This is a construction of identities by proved operations, not a search for solutions to unspecified rank equations. Ordinary block recursion on the resulting seed has exponent $\log_n R_n < 9/4 + \varepsilon$; its fixed linear maps cost quadratically per node. □

Every rational seed identity also holds over every field of characteristic zero. The preprocessing time of this procedure is finite for each fixed ε. No quantitative bound on preprocessing time or coefficient bit size is asserted.

8 Finite examples and the scope of the size bounds

The explicit family in Theorem 7.5 proves convergence with every finite charge retained. Its constants are conservative, and its guaranteed indices are far beyond useful sizes. Smaller fixed indices can be evaluated directly from their rounded counts. This section gives certified finite examples and explains their verification by exact-arithmetic enclosures.

8.1 Bounds for evaluating finite types

The entropy bounds in Section 7 can be sharpened without evaluating large factorials. Put $c_0 = (1 + \log 2)/2$. For every integer $t \geq 1$,

$$(t + \tfrac{1}{2}) \log t - t + c_0 \leq \log(t!) \leq (t + \tfrac{1}{2}) \log t - t + 1. \tag{51}$$

For the upper bound, the trapezoidal sum of the concave function $\log x$ on $[1, t]$ is at most its integral. For the lower bound, Jensen's inequality on each interval $[i - 1/2, i + 1/2]$ bounds its integral by $\log i$. Sum over $i = 1, \dots, t$, evaluate the integral, and use $(t+1/2)\log(1+1/(2t)) \geq 1/2$, a consequence of $\log(1 + x) \geq x/(1 + x)$. This proves both inequalities.

For an action with s positive child counts, let

$$C_i = \frac{1}{2} \log(N w_i) - \frac{1}{2} \sum_{y : n_{i,y} > 0} \log(N n_{i,y}).$$

Subtracting the factorial bounds gives

$$NH_i + C_i + c_0 - s \le \log M_i \le NH_i + C_i + 1 - sc_0. \tag{52}$$

Intersect these bounds with Lemma 7.4 and with $\log M_i \ge 0$. A type with one positive child count has $\log M_i = 0$ exactly. For a reset apply (52) to the two counts $(m_r, Nz_r - m_r)$, omitting any zero count, then add $(Nz_r - m_r)\log 2$. Intersect the result with $[Nz_r \log 3 - \log(Nz_r + 1),\ Nz_r \log 3]$. This uses the actual modal integer m_r, without replacing it by $Nz_r/3$.

These sector bounds give a lower bound for Δ_N and an upper bound for $\log E$. Together with the actual signed catalyst in (40) and $\log(S/H) < 3I \log 2$, they give a sufficient upper bound for Φ. All expressions involve logarithms of integer counts and small rank bases. They do not require the factorials, sector words, or coefficient arrays to be materialized.

8.2 Saved finite certificates

Table 1 records three fixed comparisons at the larger rounding scale $K = a^3\lceil \log_2 a \rceil^2$, $N = 1$, $k = 32$, and $j = 128a\lceil \log_2 a \rceil$. Increasing j amortizes catalyst supply; increasing k amortizes interpolation. Both also increase the source dimension $A = u^{kj}$.

a	K	j	upper bound for Φ	decimal digits of A
4	256	1 024	3.628926097	84 456 434
8	4 608	3 072	2.848599165	11 877 489 885
12	27 648	6 144	2.675820359	234 905 651 862

Table 1: Certified sufficient exponent bounds at fixed small indices. Here $N = 1$ and $k = 32$. Each exponent is rounded upwards from its rational interval. The digit counts refer to the recursive source side, not to a stored or executed matrix.

The last row proves that this finite relative scheme has pooled exponent less than 2.675820359. Its size also shows why this is a mathematical construction rather than an implementation result. The table does not bound the exact exponents from below, compare practical running times, or give the dimensions of the final rational seeds.

The saved certificates use rational logarithm enclosures. For a positive integer write it as $2^e y$, $1 \le y \le 2$, and bracket y by adjacent rationals of denominator 2^{80}. For $z = (y-1)/(y+1)$,

$$2\sum_{h=0}^{39} \frac{z^{2h+1}}{2h+1} \le \log y \le 2\sum_{h=0}^{39} \frac{z^{2h+1}}{2h+1} + \frac{2z^{81}}{81(1-z^2)}. \tag{53}$$

The positive tail of the power series gives the remainder bound. Evaluate at the rational bracket endpoints, apply the same bound to $\log 2$, and combine the intervals with outward rounding. For a positive rational, subtract the denominator's logarithm interval from the numerator's. Digit counts follow by enclosing $\log_{10} A$ strictly between consecutive integers. The supplementary report `results/multiway-finite-costs.json` contains the integer types, signed catalyst counts, charge intervals, and source hashes underlying the table.

8.3 Comparison with exact normalization

Appendix A gives a closed binary certificate and its exact minimal denominator K_a. It satisfies

$$K_a \ge \frac{3}{2}\operatorname{lcm}(1, \dots, a-1) \ge \frac{3\,2^{a-2}}{a}.$$

This is an obstruction to using those exact rational frequencies at a smaller scale. Conserved rounding avoids that requirement: its edge counts need only approximate the frequencies while preserving the entire resource identity. The bound $\mathcal{L} \le 4J_a$ measures the resulting loss. Multiway relations also reduce the ideal total auxiliary charge to $O(a)$. Together these changes give the bound $\log A = O(a^5(\log a)^2)$.

Threefold balancing separately reduces the dimension charges relative to sixfold balancing at the same k. It does not by itself halve an exponent: doubling k in the threefold construction recovers the same charged scalar counts as using k with sixfold balancing, when the underlying types agree. Likewise, increasing a sector count improves the target dimension but also increases the separator degree. The full expression (42) is needed to assess either change.

The dimension A belongs to the specified relative scheme. It is not a lower bound on all algorithms obtainable by restricting its source, or on all rational seeds with the same exponent. For example, tensoring a small scheme with M_C increases its displayed source side, while restricting that common factor to one coefficient recovers the small scheme. A source-size obstruction therefore needs a stated construction and reduction model. Neither the denominator benchmark nor the displayed finite examples imply a general size barrier for fast matrix multiplication.

9 Consequences and scope

The construction gives a finite answer to the extraction problem for every fixed positive exponent allowance. Its relative schemes are specified by coordinate maps and integer parameters, and their recursive use includes all finite overheads. Unrolling one fixed scheme and restricting its coefficients to the rationals gives the seed in Theorem 1.1. The argument uses the finite convolution relations from OpenAI [14], but does not assume that preprint's spectral conclusion. The theorem therefore provides a separate finite derivation of $\omega \le 9/4$.

The quantitative family also resolves a denominator problem in exact certificate constructions. Clearing the binary certificate's prescribed rational denominators requires an exponentially growing scale. That requirement belongs to those exact weights. Conserved rounding uses any positive integer scale, preserves every tensor-factor balance, and loses at most $4J_a = O(a^3)$ in the entropy expression. The resulting family has $\log A = O(a^5(\log a)^2)$, with explicit constants and a proved choice of a for every allowance. The binary denominator formula remains a useful comparison, but it is not a size barrier for this construction or for matrix multiplication algorithms in general.

The least-profile theorem identifies the exact limit of the stated scalar axioms. The diagonal lower bound follows from tripling at the row index, which is the subfamily used by the tensor maps; off-diagonal minimality uses shifted tripling beyond the row index. Since the displayed profile itself satisfies the axioms, those axioms alone cannot force faster diagonal growth. The multiway relations improve the finite accounting while keeping the same limiting profile exponent. A smaller asymptotic exponent would require additional mathematical input.

Several components can also be used separately. The determinant map turns a filtration preserved by multiplication into exact fixed-type restrictions. The occupation estimates control both the total rational multiplicity of the relations and the logarithmic cost of their auxiliaries. The circulation rounding preserves integer resource balances, while its entropy estimate remains valid for arbitrarily small or zero occupations. The separator applies to any matched sectors sharing a leg. Each use in another tensor argument would still need its own compatibility and cost analysis.

The threefold balance and pooled recursion have precise, limited effects. Alternating row orientations make three cyclic copies sufficient and reduce the dimensions at fixed parameters. This alone gives no improvement in the exponent: threefold power $2k$ reproduces the charged scalar

counts of sixfold power k for the same types and reuse count. Pooling the recursive subproblems removes the ceiling in the branch ratio, replacing $\lceil r/m \rceil$ by r/m in the exponent. It does not remove interpolation evaluations, catalyst supply, or linear transformations.

The scale bound applies to the relative source dimension A. It does not bound the dimension of the final rational seed, the size of its coefficients, or the time needed to write them out. The seed procedure terminates because it uses a fixed coefficient field and a strict gap between the constructed exponent and the requested allowance. No new small-rank identity is claimed. Every finite member of the prescribed family has exponent strictly above $9/4$, and the proof supplies no fixed recursion with cost $O(n^{9/4})$.

The allowance and coefficient set are fixed before the input dimension grows. Although Theorem 7.5 gives a rate in the profile index, it is not a construction-time or bit-complexity bound. A uniform algorithm that changes its allowance with the input dimension would require conventions for generating and loading its rational constants and a bound on that setup cost. Exact arithmetic also gives no numerical stability guarantee. The finite examples do not establish a useful matrix-size range.

Further reductions in finite cost can be assessed through the separate terms of the exponent formula. A smaller separation degree would reduce interpolation; a cheaper decomposition of the retained factors would reduce catalyst supply; and different transition rules might reduce the occupation and type costs. Such changes would need explicit maps or proved estimates. The present analysis supplies no general lower bound that excludes them.

Supporting calculations and source files

The proof is contained in the formulas and arguments in this paper. The accompanying supplement, `finite-extraction-supplement-v2.zip` (version 2, 7 October 2026), includes exact-arithmetic scripts, component checks, and the saved interval certificates used for the finite examples. The determinant and cyclic maps are checked in `experiments/multiway_determinant.py` and `experiments/cyclic_profile_extraction.py`. The occupation and rounded counts are generated by `experiments/multiway_profile_flows.py` and `experiments/multiway_rounded_flow.py`. The finite-cost evaluator is `experiments/multiway_finite_costs.py`, with its saved report in `results/multiway-finite-costs.json`. The supplement also retains the binary-certificate calculations underlying the appendix comparisons.

These files are supporting checks, not assumptions of the main theorem. The composed coefficient maps are specified mathematically; their full arrays are not stored. The reported finite bounds come from saved rational interval calculations, not from timing or executing the represented large matrix algorithms. No numerical benchmark is used to infer an arithmetic exponent.

A The exact binary certificate and its denominator scale

This appendix records a closed certificate that uses the neighboring-length relations with their exact tangent probabilities. It gives a useful comparison with the conserved rounding in Section 6. The exponential denominator bound below concerns this exact certificate; it is not a lower bound for rounded or multiway constructions.

Throughout this appendix, K, U, T, the split counts, and the catalyst refer only to the exact binary certificate. They are distinct from the rounded quantities in the main construction.

A.1 A certificate with fixed rational probabilities

For a fixed target index $a \geq 2$, use rows $2 \leq r \leq a$ and columns $r \leq b \leq 4r-2$. The affine formula in (11) holds in row r throughout $b \geq r-1$: at its left endpoint this follows from $s_r/s_{r-1} = (3r-2)/(3r-3)$. Set

$$q^{\pm}_{r,b} = \frac{2b+r-1 \pm 2}{2(2b+r-1)}. \tag{54}$$

These positive rational probabilities equal the neighboring values of F divided by $2F(r,b)$. Put $y_{r,b} = \log(P(r,b)/F(r,b))$. Weighted arithmetic–geometric mean applied to (9) yields

$$y_{r,b} - q^{-}_{r,b} y_{r,b-1} - q^{+}_{r,b} y_{r,b+1} \geq 0. \tag{55}$$

Indeed the normalized concavity inequality is

$$\frac{P(r,b)}{F(r,b)} \geq q^{-}_{r,b} \frac{P(r,b-1)}{F(r,b-1)} + q^{+}_{r,b} \frac{P(r,b+1)}{F(r,b+1)} \geq \exp\bigl(q^{-}_{r,b} y_{r,b-1} + q^{+}_{r,b} y_{r,b+1}\bigr).$$

We will also use the equality $H(q) + \sum q^{\pm} \log F(r, b \pm 1) = \log(2F(r,b))$, where $H(q) = -\sum q^{\pm} \log q^{\pm}$. Tripling at $h = r$ gives $y_{r,4r-1} - y_{r,r} \geq 0$.

Put $L = 4r-1$ and $D_r = 3r-1$. In the final row $r = a$, set

$$\lambda_{r,b} = \frac{2(2b+r-1)(L-b)}{3(r-1)D_r}, \qquad \mu_r = \frac{1}{r-1}. \tag{56}$$

In each lower row $r < a$, set

$$\begin{aligned} \lambda_{r,r} &= \frac{2D_r}{3(r-1)}, \\ \lambda_{r,b} &= \frac{4(2b+r-1)(L-b)}{3(r-1)(3r+1)} \quad (b > r), \\ \mu_r &= \frac{2D_r}{(r-1)(3r+1)}. \end{aligned} \tag{57}$$

Proposition A.1 (Rational telescoping)**.** *Weight (55) by $\lambda_{r,b}$ and tripling by μ_r. Their sum over all specified rows and columns is exactly $y_{a,a} - y_{1,2} \geq 0$. All weights and all split weights $\lambda q^{\pm}$, $\mu/3$ are positive rationals.*

Proof. Write $h_b = 2b + r - 1$ and $f_b = \lambda_{r,b}/h_b$. At an interior column the concavity contribution is $h_b(f_b - (f_{b-1} + f_{b+1})/2)$. In the final row $f_b = 2(L-b)/(3(r-1)D_r)$ is affine. The left external coefficient is $-h_{r-1} f_r/2 = -1$. At r, the concavity contribution is $r/(r-1)$; subtracting $\mu_r = 1/(r-1)$ leaves 1. The terminal coefficient is $-h_L f_{L-1}/2 + \mu_r = 0$. Thus this row leaves $y_{r,r} - y_{r,r-1}$.

In a lower row, f_b is affine for $b > r$ and $f_r = 2/(3(r-1))$. Its affine extension from $b > r$ differs at r by $2/(3r+1)$. The second difference therefore leaves coefficient 1 at $r+1$. The left external coefficient remains -1; the concavity coefficient at r is $2(3r-1)/((r-1)(3r+1)) = \mu_r$, which cancels with $-\mu_r$, and the terminal coefficient cancels with μ_r. This row leaves $y_{r,r+1} - y_{r,r-1}$. Symmetry identifies $y_{r,r+1} = y_{r+1,r}$, so all row boundaries cancel except $y_{a,a} - y_{2,1} = y_{a,a} - y_{1,2}$. □

The probabilities in this certificate are fixed by F, not by the unknown values of P. This fact is what permits an exact finite tensor construction.

A.2 Integer types and their entropy identity

Let $K = K_a$ be the least common multiple of the denominators of all λq^+, λq^-, and $\mu/3$, in lowest terms. For each concavity relation write

$$w = K\lambda, \qquad u = K\lambda q^+, \qquad d = K\lambda q^-; \qquad w = u + d.$$

For each tripling relation write $z = K\mu$; then $3 \mid z$. Let

$$U = \sum w, \quad T = \sum z, \quad I_c = \frac{3a^2 + a - 4}{2}, \quad I_t = a - 1, \quad I = \frac{3(a+2)(a-1)}{2}. \tag{58}$$

The sums run over concavity and tripling relations, respectively. There are I_c and I_t such relations.

For a formal symbol $[r, b] = [b, r]$, Proposition A.1 gives the exact integer identity

$$\sum_{r,b} (w[r,b] - u[r,b+1] - d[r,b-1]) + \sum_r z([r,4r-1] - [r,r]) = K[a,a] - K[1,2]. \tag{59}$$

This is an identity of exponent vectors, not an identity of sums of tensors. Let $i_{r,b}$ and $o_{r,b}$ be the nonnegative source and output counts on its left, after identifying symmetric pairs. Put $c_{r,b} = \min(i_{r,b}, o_{r,b})$ and

$$\mathcal{C}^{\mathrm{ex}} = \bigotimes_{r \le b} Q_{r,b}^{\otimes c_{r,b}}, \qquad \mathcal{C}_N^{\mathrm{ex}} = (\mathcal{C}^{\mathrm{ex}})^{\otimes N}. \tag{60}$$

Thus $\sum c_{r,b} = U + T - K$. The common factors are retained as the catalyst $\mathcal{C}^{\mathrm{ex}}$; they are not formally canceled from a restriction.

Tightness of each entropy relation at F also gives

$$2^{K-U} 3^T \prod_{\text{concavity relations}} \frac{w^w}{u^u d^d} = F(a,a)^K. \tag{61}$$

To check the identity, multiply $(2F(r,b))^w = (w^w/(u^u d^d))F(r,b+1)^u F(r,b-1)^d$ and $F(r,4r-1)^z = 3^z F(r,r)^z$ over the relations. Use (59) and $F(1,2) = 2$. This derivation proves the identity over the rationals, without a numerical entropy calculation.

A.3 The minimal denominator scale

Theorem A.2 (Exact normalization)**.** *The least integer scale in Section A.2 is*

$$K_a = \operatorname{lcm}\left(\frac{3(a-1)(3a-1)}{\gcd(3a-1,4)}, \quad \left\{ \frac{3t(3t+4)}{\gcd(t,2)^2} : 1 \le t \le a-2 \right\} \right). \tag{62}$$

In particular,

$$3\operatorname{lcm}(1,\dots,a-1) \mid 2K_a, \qquad K_a \mid 3\operatorname{lcm}(1,\dots,3a-1), \tag{63}$$

and

$$K_a \ge \frac{3}{2}\operatorname{lcm}(1,\dots,a-1) \ge \frac{3\,2^{a-2}}{a}. \tag{64}$$

Proof. In the final row put $t = a-1$, $D_r = 3t+2$, and $b = a+h$. All split weights have common denominator $3tD_r$. Their integer numerators are D_r, from $\mu/3$, and $(D_r + 2h \pm 2)(D_r - h)$ for $0 \le h \le D_r - 1$. Modulo D_r, the latter products are $-2h(h \pm 1)$, all divisible by four. The case $h = 1$ with the plus sign is congruent to -4. Thus the joint gcd of the denominator and all numerators is exactly $\gcd(D_r, 4)$. This gives the first entry in (62).

In a lower row put $t = r - 1$ and $V = 3t + 4$. Use common denominator $3tV$. At $b = r$, the two split numerators are V^2 and $3tV$; the tripling split numerator is $2(3t + 2)$. For $b = r + h > r$, the split numerators are $2(3t + 2 + 2h \pm 2)(3t + 2 - h)$. An odd prime dividing both V^2 and $2(3t + 2)$ would divide the difference between V and $3t + 2$, namely two, which is impossible. If t is odd, V^2 is odd, so the joint gcd is one. If t is even, all numerators and the denominator are divisible by four. When $t \equiv 2 \pmod 4$, V^2 has exactly two factors of two; when $t \equiv 0 \pmod 4$, $2(3t + 2)$ has exactly two. Thus the joint gcd is exactly four. This proves the remaining entries.

Each row denominator is either $3\,\mathrm{lcm}(t, 3t + c)$ or half that number, with $c = 2$ for the final row and $c = 4$ for lower rows. The integers involved lie between one and $3a - 1$; twice each row denominator is divisible by $3t$. Taking least common multiples proves (63).

For completeness, every binomial coefficient $\binom{n}{h}$ divides $\mathrm{lcm}(1, \dots, n)$. In its factorial valuation at a prime p, each power $p^j \le n$ contributes zero or one, so its valuation is at most $\lfloor \log_p n \rfloor$. The largest binomial coefficient in row n is at least $2^n/(n + 1)$. Apply this with $n = a - 1$ and use (63) to obtain (64). □

The scale is minimal for these exact rational split weights. Approximate types, different weights, or different tensor maps need not obey this bound. Since $U \ge K_a$, already the sixfold binary source dimension 2^{2NU} grows at least doubly exponentially in a under this bound. The inequality $U \ge K_a$ also follows directly from (59): the net consumption of $[a, a]$ is K_a, and all its source occurrences belong to concavity relations.

B The full composed relation maps

This appendix specifies the composition in (39) on its full variable spaces. In particular, the sector-specific reindexing is a fixed linear map even on positive-degree terms, whose first-leg label need not agree with the other sector labels.

For a split action $i \in \mathcal{I}_c$, write its row, source column, and width as r_i, b_i, h_i, and put $d_i = h_i + 1$. Let w_i be its source count and $n_{i,q}$ its child counts, ordered by decreasing length $b_i + h_i - 2q$, $0 \le q \le h_i$. Thus $\sum_q n_{i,q} = w_i$. For a tripling relation in row r_i, use $w_i = z_{r_i}$ in the formulas below. At type scale N, put $\nu_i = Nw_i$; in the tripling case also put $m_i = \lfloor (\nu_i + 1)/3 \rfloor$. The source and canonical sector tensors are

$$\mathcal{U}_i = \begin{cases} (C(r_i, b_i) B_X(d_i))^{\otimes \nu_i}, & i \in \mathcal{I}_c, \\ C(r_i, 4r_i - 1)^{\otimes \nu_i}, & i \in \mathcal{I}_t, \end{cases}$$

and

$$\mathcal{V}_i = \begin{cases} \displaystyle\prod_{q=0}^{h_i} C(r_i, b_i + h_i - 2q)^{\otimes N n_{i,q}}, & i \in \mathcal{I}_c, \\ C(r_i, r_i)^{\otimes(\nu_i - m_i)} (C(r_i, r_i)^{(23)})^{\otimes m_i}, & i \in \mathcal{I}_t. \end{cases}$$

The tripling sector is kept in this mixed orientation until the three permutations are combined.

Let T_i be the exact type map of Lemma 4.1 or Lemma 4.2. Enumerate its retained words lexicographically by $s = 0, \dots, M_i - 1$, and write

$$\mathsf{T}_i(\mathcal{U}_i) = \sum_{s=0}^{M_i - 1} \sum_{\alpha, \beta, \gamma} c_{i,s}(\alpha, \beta, \gamma) x_\alpha y_{s,\beta} z_{s,\gamma}.$$

The first variable space X_i is shared. The other two spaces are $\bigoplus_s Y_{i,s}$ and $\bigoplus_s Z_{i,s}$, and the indices β, γ range over the coordinates of sector s.

Set $L_i = 2M_i - 1$, and let $\mathsf{S}_i(t)$ be the substitutions of Proposition 4.3 with $K_0 = M_i, c = 1$. Applied after the type maps in the L_i source copies, they give exactly

$$\mathsf{S}_i(t)\left(\bigoplus_{\ell=0}^{L_i-1}\mathsf{T}_i\right)(L_i\mathcal{U}_i) = \sum_{\substack{0\le s,g,u,v<M_i\\ u-v+g-s=0}} t^{(g-s)^2}\sum_{\alpha,\beta,\gamma} c_{i,s}(\alpha,\beta,\gamma)X_{\alpha,g}Y_{s,\beta,u}Z_{s,\gamma,v}. \tag{65}$$

This formula includes every positive-degree term. The Fourier sum imposes the equality $u-v+g-s = 0$, rather than merely a congruence, and the combined weight is $(g-s)^2$.

Here the coefficients are explicit integer arrays. For a split action, let $\omega_i(s) \in \{0,\dots,h_i\}^{\nu_i}$ be its retained word. In the determinant bases, before reordering positions,

$$c_{i,s}(\boldsymbol{\alpha},\boldsymbol{\beta},\boldsymbol{\gamma}) = \prod_{\theta=1}^{\nu_i}\mathbf{1}_{\{\gamma_\theta=\alpha_\theta+\beta_\theta\}}.$$

At a position with label $q = \omega_i(s)_\theta$, the coordinate ranges are

$$0\le\alpha_\theta<r_i,\quad 0\le\beta_\theta<b_i+h_i-2q,\quad 0\le\gamma_\theta<r_i+b_i+h_i-2q-1.$$

Thus the coefficient of $X_{\boldsymbol{\alpha},g}Y_{s,\boldsymbol{\beta},u}Z_{s,\boldsymbol{\gamma},v}$ in (65) is

$$\mathbf{1}_{\{u-v+g-s=0\}}\,t^{(g-s)^2}\prod_{\theta=1}^{\nu_i}\mathbf{1}_{\{\gamma_\theta=\alpha_\theta+\beta_\theta\}}.$$

For a tripling relation, the same product of indicators applies when $\beta_\theta,\gamma_\theta$ are the original interval coordinates from Lemma 4.2. The word-specific shifts and first-index reversals are part of the subsequent map, not this coefficient array.

The separator target spaces are

$$\widetilde{X}_i = \bigoplus_g X_i^{(g)},\qquad \widetilde{Y}_i = \bigoplus_s (Y_{i,s}\otimes\mathbb{C}^{M_i}),\qquad \widetilde{Z}_i = \bigoplus_s (Z_{i,s}\otimes\mathbb{C}^{M_i}).$$

For each word s, let $(\mathsf{R}^X_{i,s},\mathsf{R}^Y_{i,s},\mathsf{R}^Z_{i,s})$ be the coordinate permutations identifying its sector tensor with $\mathcal{V}_i$. For a split, they order positions by increasing filtration index, preserving the original order within each class. For tripling, they place all outer positions before all middle positions, again preserving their order; they shift the interval coordinates and reverse the first index at middle positions. Both outer labels have the same canonical block dimensions, so $\mathcal{V}_i$ is independent of the numbers of left and right labels.

Define the reindexing on the full target spaces by

$$\mathsf{R}_i = \left(\bigoplus_g \mathsf{R}^X_{i,g},\quad \bigoplus_s(\mathsf{R}^Y_{i,s}\otimes I_{M_i}),\quad \bigoplus_s(\mathsf{R}^Z_{i,s}\otimes I_{M_i})\right).$$

The first component uses the word indexed by g, even in a monomial whose other sector label is $s \neq g$. This defines a coordinate permutation on every target variable. Its behavior on those positive-degree monomials need not identify them with sector tensors; only the constant term has that form. All three maps are independent of t.

Choose the row sign $\sigma_i = (-1)^{a-r_i}$, and let $\pi \in \mathcal{A}_{\sigma_i}$. The complete map $\Psi_{i,\pi}(t)$ acts on $L_i\mathcal{U}_i^\pi$; its separator uses the global leg occupied by the native first leg after permutation by π. The constant term in (65) has $g = s$ and $u = v$, so

$$\Psi_{i,\pi}(t)(L_i\mathcal{U}_i^\pi) = \bigoplus_{s=0}^{M_i-1} (\mathcal{V}_i B_X(M_i))^\pi + \sum_{e=1}^{(M_i-1)^2} t^e G_{i,\pi,e}.$$

The first term is a full direct sum on all three legs.

Now tensor over the three $\pi \in \mathcal{A}_{\sigma_i}$. A direct-sum block is indexed by a triple of independently chosen sector labels. In every such block the canonical tensor is the same product $\bigotimes_\pi (\mathcal{V}_i B_X(M_i))^\pi$. Regrouping its factors balances the dot products into M_{M_i}. For tripling, apply (17) to its convolution factors. This is a fixed coordinate isomorphism of the canonical product spaces, applied separately on every block of each global leg. In particular it is also defined on positive-degree monomials whose three block labels differ. It introduces no degree or interpolation cost.

Finally tensor over all actions, using one parameter. The source and constant-term multiplicities are

$$\prod_i L_i^3 = S, \qquad \prod_i M_i^3 = H,$$

and Lemma 2.1 bounds the degree by $3\sum_i (M_i - 1)^2 = D$. The signed ledger identifies common source and target convolution factors by the actual coordinate isomorphisms $S_{r,b}^\sigma \cong S_{b,r}^{-\sigma}$, then retains them as the catalyst. Combining the auxiliary factors by (1) gives precisely (38). All final regroupings are fixed maps on the full spaces, not instructions defined only for the constant term.

C Sparse exactification and a two-power refinement

These refinements explain which costs of a separator can be reduced. The general extraction in Section 7 uses the simpler degree bound of Lemma 2.2. If $G(t) = \sum_{s\in E} t^s G_s$, with finite nonnegative integer support E containing zero, define

$$P(z) = \prod_{s\in E\setminus\{0\}} \frac{z - 2^s}{1 - 2^s} = \sum_{j=0}^{|E|-1} \alpha_j z^j.$$

Then $\sum_j \alpha_j G(2^j) = G_0$. This gives an exact rational interpolation formula using $|E|$ evaluations. For a separator, each evaluation uses L source copies, for a total of $L|E|$. For one separator, applying this prescription to the guaranteed support set $\{cd^2 : 0 \le d \le d_*\}$ cannot improve on direct copying: the prescribed count satisfies

$$L(d_* + 1) \ge MK_0. \tag{66}$$

If $d_* = M - 1$, use $L \ge K_0$. Otherwise $d_* + 1 = \lceil K_0/c \rceil$, and $L(d_* + 1) \ge K_0(d_* + 1) + K_0(M - 1) \ge MK_0$. Direct copying uses one source for each pair (h, u), and restricts that source to sector h. Equation (66) concerns these particular copy counts, not all possible exactifications.

A smaller exact projection is possible for two powers of the separator polynomial G. If a prime q satisfies $q > d_*$, $q \equiv 3 \pmod 4$, and $q \nmid c$, then for a primitive q-th root η,

$$\frac{1}{q}\sum_{j=0}^{q-1} G(\eta^j)^{\otimes 2} = G_0^{\otimes 2}. \tag{67}$$

Indeed a surviving exponent would require $d_1^2 + d_2^2 \equiv 0 \pmod q$. Since -1 is not a square modulo q, both d_i vanish modulo q, hence are zero. For $M = K_0 = 8, c = 1$, one can use $L = 15, q = 11$, giving

$11 \cdot 15^2 = 2475$ source copies for the squared tensor, compared with $8^4 = 4096$ by direct copying. No unbounded prime search is needed for the main result: the general degree projection is always available.

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